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# Versatile Tool for Parametric Smooth Turbomachinery Blades

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**Abstract:** Designing blades for efficient energy transfer by turning the flow and angular momentum change is both an art and iterative multidisciplinary engineering process. A robust parametric design tool with few inputs to create 3D blades for turbomachinery and rotating or non-rotating energy converters is described in this paper. The parameters include axial–radial coordinates of the leading/trailing edges, construction lines (streamlines), metal angles, thickness-to-chord ratio, standard, and user-defined airfoil type among others. Using these, 2D airfoils are created, conformally mapped to 3D stream surfaces, stacked radially with multiple options, and they are transformed to a 3D Cartesian coordinate system. Smooth changes in blade curvature are essential to ensure a smooth pressure distribution and attached flow. B-splines are used to control meanline curvature, thickness, leading edge shape, sweep-lean, and other parameters chordwise and spanwise, making the design iteration quick and easy. C2 curve continuity is achieved through parametric segments of cubic and quartic B-splines and is better than G2. New geometries using an efficient parametric scheme and minimal CAD interaction create watertight solid bodies and optional fluid domains. Several examples of ducted axial and radial turbomachinery with special airfoil shapes or otherwise, unducted rotors including propellers and wind and hydrokinetic turbines are presented to demonstrate versatility and robustness of the tool and can be easily tied to any automation chain and optimizer.

**Keywords:** parametric blade; turbomachinery; smooth C2 continuous B-spline

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## 1. Introduction

Engine performance improvement is realized from optimizing several sub-components and designing turbomachinery blade shapes to convert energy efficiently. A robust and general parametric approach to blade design is essential in shape optimization and novel geometry exploration to overcome complex flow 3D effects. The main property of a blade is to efficiently and smoothly turn flow and change fluid velocity. Thus, blade design is a very crucial part of any engine, pump, or turbine design process which is iterative in nature. Parametric design of blade shapes is the best approach for such processes as the design cycle can be repeated quickly and easily after every necessary modification to blade geometry. A common design principle is to create 2D airfoils with meanline camber, stack them radially, and generate 3D blades. BladeCAD developed by Oliver et al. [1] is an interactive tool and blade sections are created with respect to general surfaces of revolution using a non-uniform rational B-spline (NURBS) surface. Blade design is defined by constructing a mean camber line, which is a Bezier curve of third degree defined by blade angles, top and bottom curves, and the stagger angle. Kulfan et al. [2–4] demonstrated the benefits of class function/shape function approach in airfoil creation. Relatively, minimal variables are needed to create various internal and external aerodynamic bodies. Sripawadkul et al. [5] described the importance of parameterization using characteristics such as parsimony, intuitiveness, orthogonality, completeness, and flawlessness in airfoil design and optimization. Koini et al. [6] described a tool for parametric design using NURBS via control points. Usually, Bezier curves of order 3 or higher are used for the mean camber definition to avoid curvature discontinuities and NURBS are used for joining curves as it is the most widely used set of spline curves in many CAD systems. Several authors in the past have used monomial,

Bezier, B-spline, and NURBS basis functions for G1 and G2 continuous 3D blades [7–13] with some caveats. Leylek et al. [14] described the mapping from the geometric space to a parametric space in which all feasible geometric configurations can be generated within a unit hyper-cube boundary.

Korakianitis [15–17] defined curvature distribution of blade surfaces as a spline, then matched slope of curvature of blade surface with four contact points of leading and trailing edges together. Burman et al. [18] defined suction and pressure sides with a thickness distribution and camber line splines. Pressure distribution is directly dependent on the curvature according to the radial equilibrium equation with smooth thickness distribution defined. Hamakhan et al. [19] illustrated the advantages of a direct surface-curvature distribution blade-design method on leading edge shape. Song et al. [20] studied the effect of leading-edge blend point curvature and main-surface curvature continuity using RANS and LES. A continuous curvature leading edge removes the pressure spike and related separation. Bi et al. [21] developed a curvature approach for transonic centrifugal compressors using a fifth order B-spline curve by using the first order differential of the curvature with respect to arc length as the smoothness measure. Meckstroth et al. [22] presented a method to enable probabilistic design based on blade manufacturing variation using a point cloud to parameter approach to create digital shapes from scans for analysis. Similarly, McKeand et al. [23] developed a method to model uncertainty in manufactured blades through a two-phase approach of 3D blade scan and parametric model generation. Recent additive manufacturing techniques have helped designers to create novel blade designs by pushing the feasibility envelope. Blade shapes have evolved over the last few decades as demonstrated by Benini et al. [24], Lian et al. [25], and Ellbrant et al. [26]. Zhang et al. [27] used a curvature-controlled stacking line method for compressor design using Bezier splines for smooth surface. Sharma et al. [28] added a continuous modified NACA four-digit thickness distribution to the geometry generator presented in this paper. It uses second derivative parameterization for the meanline and adds a symmetric thickness distribution. They also connected the tool to an open-source CAD system [29–32] and generated user-defined primitives to create 3D solids and flends to represent fillets. Agromayor et al. [33] demonstrated a general parameterization for turbomachinery using NURBS curves and surfaces with G2 continuity and showed shape derivatives using complex-step method to calculate sensitivities. Luo et al. [34] paired the geometry generator described here with a convolutional neural network to create a blade surrogate model using deep learning techniques which is the natural way forward in design. Hengtao Shi [35,36] proposed a parametric design method for high-speed axial compressors using normalized camber angle definition and thickness distribution instead of camber line approach. Yang et al. [37] proposed a flexible method for axial compressor geometry design using camber angle definition to create camber line and added thickness distribution.

Tip leakage flows in a shrouded compressor rotor, wind turbine tip vortices and secondary flow losses are very complicated phenomenon and are hard to minimize using traditional methods of blade design. Winglets are wing tip devices that reduce losses at the tip by weakening vortex shedding and reduce downwash, which increases the effective angle of attack. This enables the blade to operate closer to the designed geometric angle of attack and increases power extraction in wind turbines [38] for example. Tip clearance flows in ducted machines can reduce pressure rise, flow range, and efficiency and is a major contributor to loss. Tip leakage flow is generated by pressure difference between the pressure side and suction side of the rotor tip and introduces a high velocity flow through the tip clearance gap which is ejected into the passage. In addition to this effect, relative motion of the end wall with respect to tip leakage flow is in the same direction as the leakage flow [28] in ducted machines and serves to enhance them. A tip gap vortex is formed as the pressure-driven leakage flow interacts with main passage flow and can interact with the end wall boundary layer, drawing low momentum fluid further into the main passage flow. Rear stages of axial compressors decrease in annulus height, blade aspect ratio, and—due to a constant clearance—a high relative clearance ratio is present

and is dominated by end wall flows [39]. Traditional method of blade design is not enough to overcome these effects. Novel design exploration is imperative and is demonstrated to improve the flow [40,41]. Kinetic energy dissipation downstream of the device can be controlled through efficient design. Spanwise Reynolds number distribution and geometry aspect ratio have a direct impact on turbulent kinetic energy and anisotropy in turbulence for rotating [42,43] and non-rotating energy converters. Choosing the right turbulence model is crucial in performance prediction at various fidelities, and Bhide et al. [44,45] demonstrated this fact by comparing Boussinesq and non-Boussinesq turbulent RANS models with LES results for a supersonic rectangular nozzle flow. Accounting for turbulent kinetic energy budget downstream of energy exchange, a device is required to understand the various contributors to loss as demonstrated by Bhide et al. [46]. Better geometrical representation enables improved energy exchange.

The geometry generator described in this work has the capability of varying parameters smoothly in chordwise and spanwise directions to create traditional and novel blade shapes which minimize tip flow and secondary flow losses, as well as improving performance. Sweep and lean capabilities are implemented locally and globally to overcome several complex flow effects as demonstrated further. Curvature defined meanline approach other than the traditional method is applicable for any blade definition and combines the best spline methods in a simple manner. It also opens the design space by parametric use of B-splines with the freedom to use any general airfoil or cross section geometry. C2 parametric continuity of curves is achieved. S-shaped airfoils for compressors are beneficial in minimizing the separation and shock losses, as well as for improving efficiency, and they are feasible using the curvature method. Radial machines are also improved by using parametric manipulation tied with an optimizer to obtain feasible novel geometries. Spanwise chord and stagger with an external airfoil shape is used to generate unducted rotors. The goal is to have a unified turbomachinery tool for ducted axial and radial machines and high-aspect-ratio unducted turbomachines. Robust and inexpensive parameterization of several geometrical inputs is crucial in making the tool versatile. This paper describes the entire procedure of converting 2D to 3D airfoils including stacking and streamwise mapping, and lean and sweep capabilities which add to the generality. A wide variety of applications—including axial compressors and turbines, radial compressors, unducted propellers, and wind and hydrokinetic turbines—are demonstrated.

## 2. Methodology

### *2.1. Overview*

A general method of designing 2D blade sections using input parameters and stacking them radially with many options to obtain a 3D blade model is demonstrated. Smooth blades are achieved through robust application of essential geometrical and aerodynamic parameters as input along with the construction line (streamline) coordinates defined by an axisymmetric run like T-Axi [47], or smooth curves defined between the hub and casing or other low-fidelity tools defining spanwise geometrical properties [48]. The input also has airfoil stacking information and a control table which defines the sweep, lean, angles, and any required quantity with a few control points of cubic B-splines for adding more definition to the blade geometry. A 3D blade is generated by radially stacking 2D sections created in $(u, v)$ space as shown in Figure 1. The data input file describes curves in $(x_s, r_s)$ space and blade parameters such as section metal angles, chord, and maximum thickness. Control parameters describe 2D section curvature, thickness modifier, and leading and trailing edge shapes. Using a CAD package, 3D blade sections are lofted into a solid 3D blade shape.

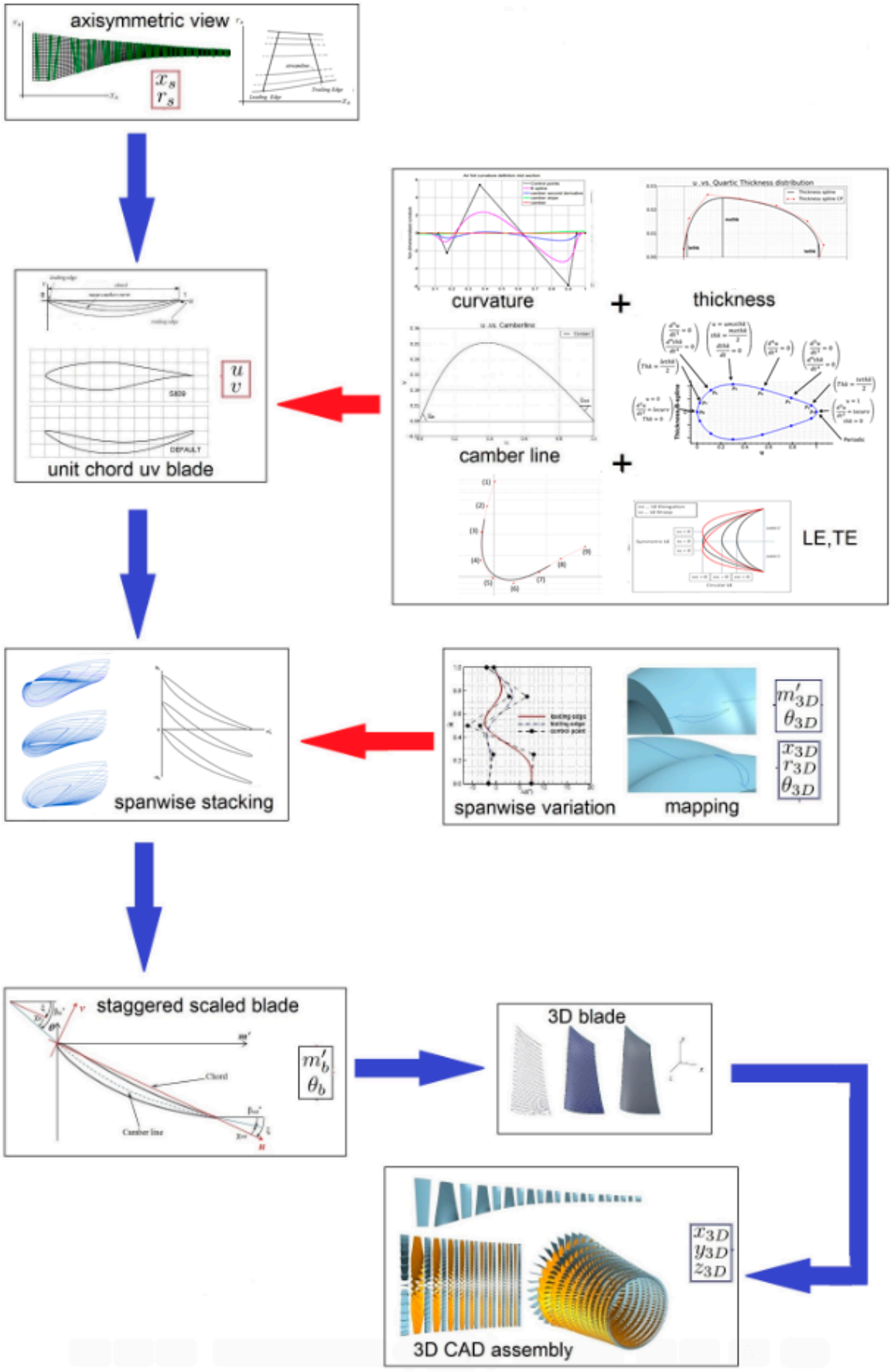



**Figure 1.** Flowchart of the entire parametric 3D blade generation process.

### 2.2. Coordinate Systems

A variety of coordinate systems are used to create a 3D blade in Cartesian coordinates from an axisymmetric $(x_s, r_s)$ system. Figure 2 shows the progression of coordinate systems used. Meridional view of streamlines with leading and trailing edges is also shown in Figure 2 for axial and radial devices. A 3D blade is constructed using streamline and airfoil coordinates. The meridional coordinate is calculated for varying and constant radius streamlines as

$$m'_s = \int \frac{dm_s}{r_s} = \frac{\sqrt{(dx_s)^2 + (dr_s)^2}}{r_s}$$

$$m'_s = \int \frac{dx_s}{r_s} = \frac{x_s}{r_s} \; (constant\ radius\ lines)$$

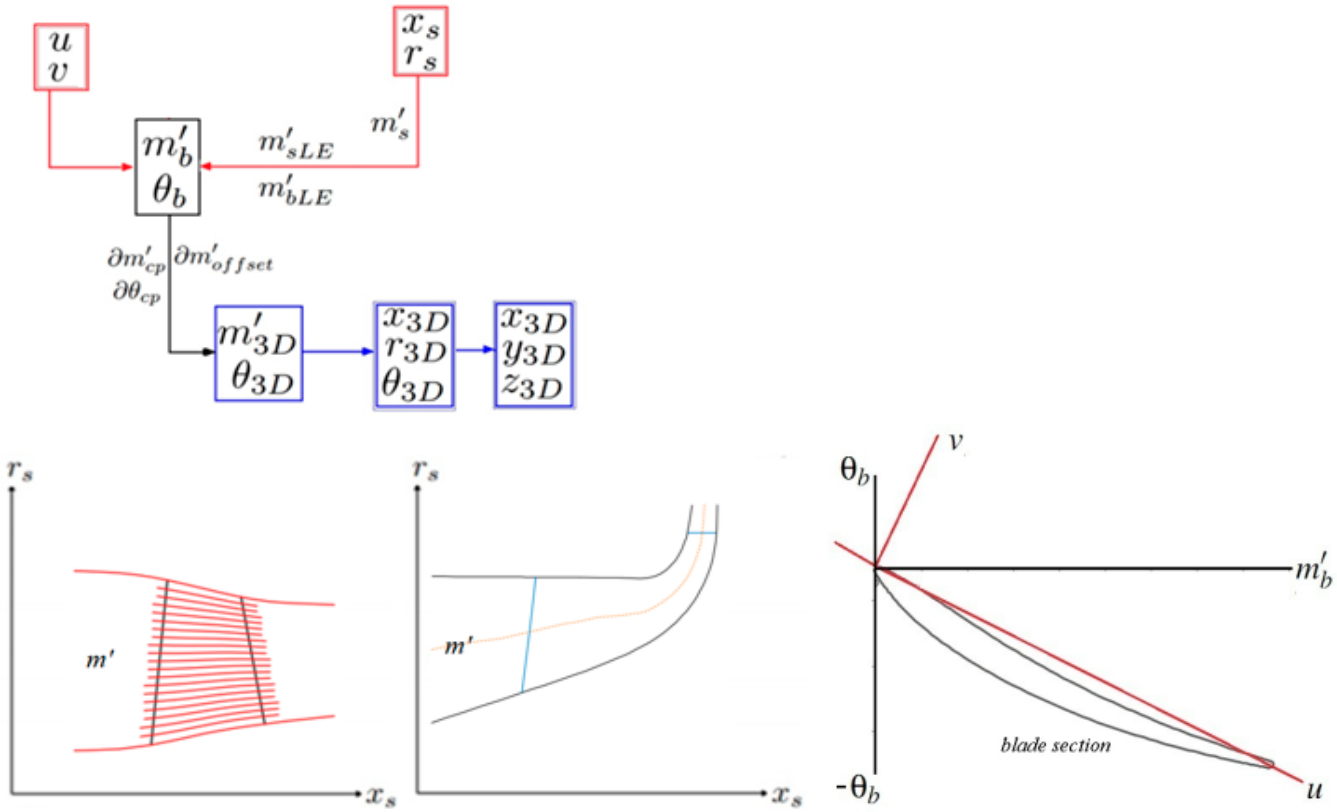



**Figure 2.** Coordinate systems used for 3D blade generation with meridional view for both configurations and relation between $(u, v)$ and $(m'_b, \theta_b)$ axes (right).

Camber-line or meanline is created in the unit chord $(u, v)$ space. Thickness distribution and leading and trailing edges are added to the camber-line to create a 2D blade section. Lean is defined as either tangential or normal to the blade chord [49] and sweep as meridional or along the chord [50]. A stacking offset in $u$ and $v$ relative to percent chord is applied to the $u$ and $v$ coordinates. The resulting 2D airfoil is rotated by stagger angle and scaled either by non-dimensional calculated chord (*chrd*) or user defined chord axial projection (*chrdx*), as shown in Figure 3. Lean and sweep are applied in this coordinate system. The blade is created in cylindrical system and normal practice is to obtain a 3D blade in Cartesian coordinates as most CAD packages use this system. Therefore, a coordinate transformation from a cylindrical to Cartesian system is necessary to obtain $x_{3D}$, $y_{3D}$, and $z_{3D}$ blade coordinates. Engine axis is assumed to be along the $X$-axis which makes x-values remain the same in both coordinate systems. $(m'_b, \theta_b)$ axis system is a scaled and rotated $(u, v)$ axis system as shown in Figure 2.

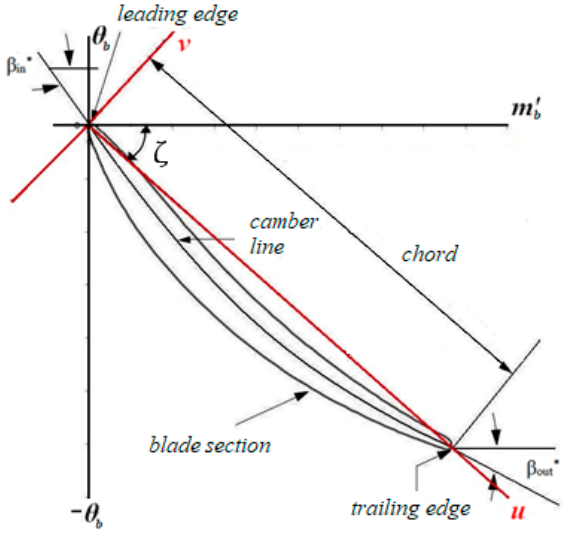



**Figure 3.** Blade section on the meanline curve with blade angles.

### *2.3. 2D Airfoil Construction*

2D blade construction uses metal angles, $\beta^*$; and sometimes flow angles, $\beta$. Often, the differences are needed. One consistent angle definition is considered for all blades (compressor and turbine) considering the sign of each angle depending on its direction, as shown in Figure 3. Both blade angles and inlet camber angle are negative for this compressor blade configuration, and the outlet camber angle is positive. Details of airfoil construction with incidence and deviation angles are explained by Siddappaji [51].

Blade creation is non-dimensional and is normalized using tip radius of the leading edge. Zero stagger 2D airfoil is created in ($u$, $v$) plane with unit chord in $u$ direction. $u$ coordinates range from 0 to 1 and are clustered near both ends using a simple sine definition. Both $u$ and $v$ coordinates can either come from an external airfoil data bank or $v(u)$ can be obtained by parametrically defining chordwise curvature, $v''(u)$ and integrating it twice. Airfoil thickness—$thk(u)$—is also defined parametrically as a function of $u$, either using Wennerstrom's third order polynomial distribution [52] or quartic B-spline distribution [53]. More recently, a modified NACA four-digit distribution is added by Sharma et al. [54]. As a collaborative research tool originated at University of Cincinnati more than a decade ago [55], features are always updated, and new capabilities are added by current researchers to improve the robustness and versatility.

Top and bottom airfoil curve coordinates are generated using the equations below which include thickness definition, camber, and meanline coordinates. 2D stacking procedure aligns the airfoil to stacking coordinates $u_{stack}$, $v_{stack}$. Top and bottom curves are joined at LE and TE to create $u_b$, $v_b$ coordinates which are rotated by a stagger angle of $\zeta$ and scaled up by non-dimensional chord obtained as shown below. These 2D ($m_b'$, $\theta_b$) coordinates are converted into streamwise mapped coordinates ($m_{3D}'$, $\theta_{3D}$) by including a meridional offset for accurate conformal mapping. Meridional streamline coordinate, $m_s'$, is constructed from $x_s$ and $r_s$ coordinates. $m_{sLE}'$ is the streamline LE meridional coordinate obtained by inverse spline of $x(m_s')$ evaluated at $x_{LE}$ on each streamline for axial LE or inverse spline of $r(m_s')$ evaluated at $r_{LE}$ on each streamline for radial LE. Sweep and lean perturbation are also embedded in these coordinates. $m_{sLEhub}'$ is added when sweep perturbation is defined to avoid wiggles at LE. It is the $m_{sLE}'$ at the LE of hub airfoil. $x_{3D}$ and $r_{3D}$ coordinates are obtained by evaluating splines at each $m_{3D}'$ for all the defined points. Cartesian coordinate transformation generates 3D blade coordinates and they are scaled by the dimensional blade scaling factor ($scf$) for CAD creation with $X$-axis as the engine axis.

$$
\begin{aligned}
u_{bot} &= u + (thk)\sin\left(tan^{-1}v'(u)\right) \\
v_{bot} &= v(u) - (thk)\cos\left(tan^{-1}v'(u)\right) \\
u_{top} &= u - (thk)\sin\left(tan^{-1}v'(u)\right) \\
v_{top} &= v(u) + (thk)\sin\left(tan^{-1}v'(u)\right) \\
u_{b_bot} &= u_{bot} - u_{stack} \\
v_{b_bot} &= v_{bot} - v_{stack} \\
u_{b_top} &= u_{top} - u_{stack} \\
v_{b_top} &= v_{top} - v_{stack} \\
chrdx &= \left|m'_{sTE} - m'_{sLE}\right| \\
chrd &= chrdx/\left|\cos(\zeta)\right| \\
m'_b &= chrd(\, u_b\cos(-\zeta) + v_b\sin(-\zeta)) \\
\theta_b &= chrd(-u_b\sin(-\zeta) + v_b\cos(-\zeta)) \\
m'_{3D} &= m'_b + \delta m'_{offset} + \delta m'_{sweep} \\
\theta_{3D} &= \theta_b + \delta\theta_{lean} \\
\delta m'_{offset} &= m'_{sLE} - m'_{bLE} \\
x_{3D} &= scf \text{ x } (x_{3D}) \\
y_{3D} &= scf \text{ x } (r_{3D}\cos(\theta_{3D})) \\
x_{3D} &= scf \text{ x } (r_{3D}\sin(\theta_{3D}))
\end{aligned}
$$

An elliptical LE and circular TE are used as default. Elliptical leading edge helps in keeping the flow attached and laminar [56]. Geometry generator has an option of choosing the shape of leading and trailing edges. Default camber line is constructed using the blade angles ($\beta^*_{in}$, $\beta^*_{out}$), thickness to chord ratio, and meridional chord value. A mixed camber line is defined which is partly cubic in nature as default. The relationship between the ($m'_b$, $\theta_b$) and ($u$, $v$) system is shown in the Figure 2. The top and bottom airfoil curves are then calculated using the camber line, thickness distribution, and blade angles at each coordinate point. The equations for the default camber curve and blade coordinate generation are explained by Siddappaji [51]. Parametric cubic [57] and quartic [58,59] B-splines are used to create curvature, thickness distribution, and thickness modifier for 2D sections using control points. Furthermore, spanwise distribution of several properties are defined using cubic B-spline with very few control points to construct a smooth blade. A geometry generator is flexible in generating different airfoil shapes to construct a variety of blade designs. Since each blade section is constructed independently through parameters, it is possible to have different blade sections at different spanwise locations stacked differently. For example, a wind turbine blade has a circular hub and progresses towards an S809 airfoil shape radially. Consistent and general capability of the blade section construction makes it easy to be robust with such shapes. The tool can also import external airfoils in the ($u$, $v$) plane. Curvature driven meanline definition adds more freedom to the capability. Sherar [60] describes the curvature as a function of first and second derivatives,

$$C = R^{-1}_{curv} = \frac{y''}{\left(1 + y'^2\right)^{3/2}}$$

The second derivative of the meanline (camber line) is defined as a cubic B-spline. Figure 4 shows second derivative B-spline representing curvature of the blade section from leading edge to trailing edge (from $u$ = 0 to $u$ = 1) and the meanline which is obtained through an iterative process of integrating curvature and the slope curve in that order. Second derivatives $v''$ are integrated instead of curvature to eliminate non-linearity of arc length [53]. The above equation shows that the second derivative has the characteristic of curvature. There are four options for thickness distribution in the tool: Wennerstrom thickness distribution (third order polynomial), Quartic B-spline thickness distribution for blunt and sharp TE, and a modified NACA four-digit distribution [28]. Another option of LE is implicitly defined in the quartic B-spline thickness distribution with appropriate constraints. Options for trailing edge are same as the leading edge with added options of a

circular arc and a blunt trailing edge. Figure 4 shows several 2D airfoils used in a geometry generator either imported or generated using a curvature-driven meanline construction process for various ducted and unducted applications.

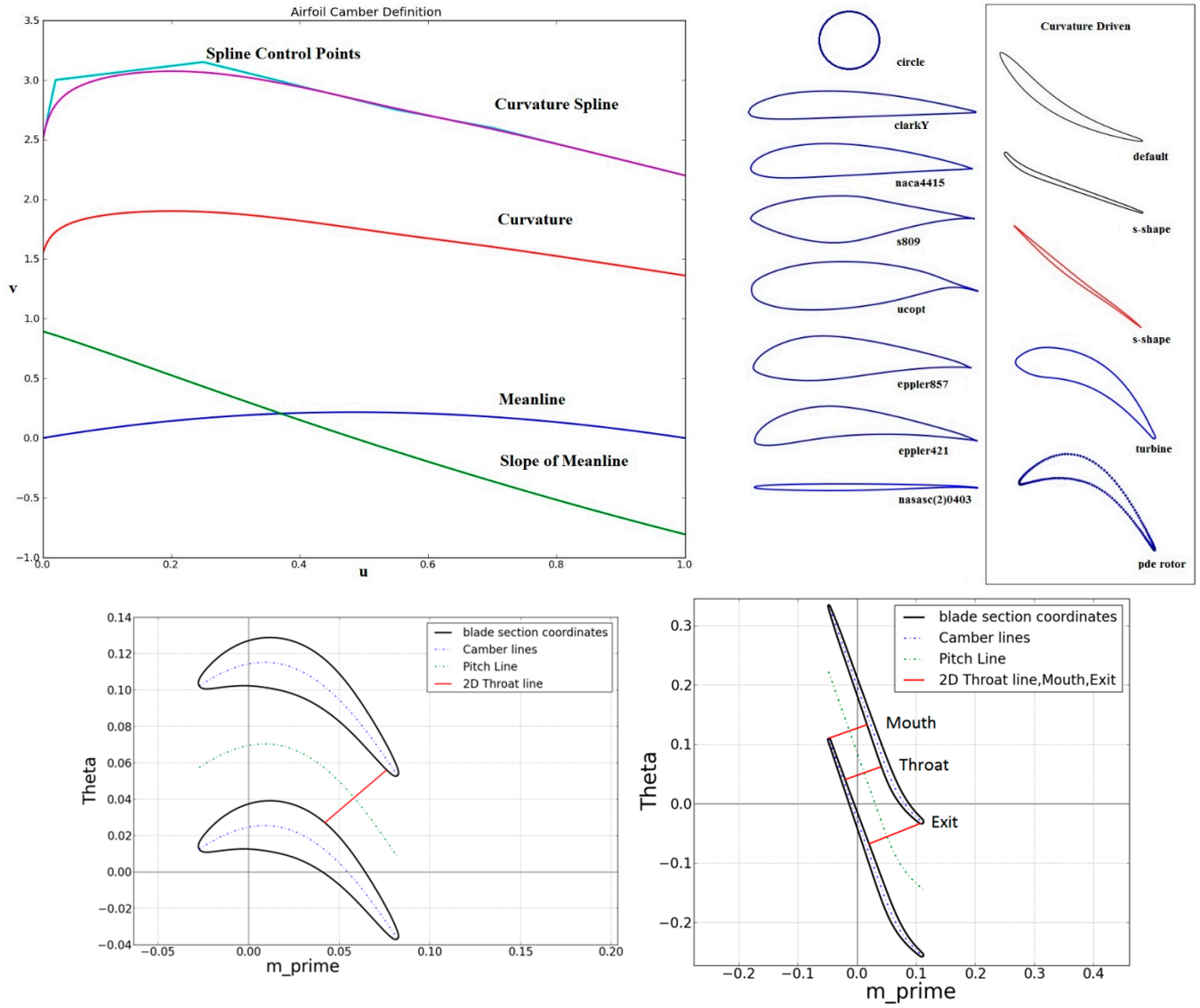



**Figure 4.** Second derivative cubic B-Spline and meanline in $(u, v)$ space. Several 2D airfoils used, either imported or created using curvature driven meanline process. 2D throat for a turbine and a transonic fan rotor cascade.

Another feature is the ability to calculate the 2D and 3D throat of the blade passage which is a crucial parameter in optimizing blockage and other related aerodynamic performance. 2D throat is defined as the minimum distance normal to the 2D pitch line and intersects the two adjacent blades. 2D pitch line is defined as an imaginary 2D mean line halfway between two adjacent sections in 2D cascade view as shown in Figure 4. This 2D throat is also mapped to the 3D coordinate system to obtain the 3D throat lines at several blade span [53]. The flowchart for generating 2D airfoils with various options implemented robustly is shown in detail by Siddappaji [48] and is described in Appendix A, Figure A1.

### *2.4. Cubic and Quartic B-Spline Implementation for Smoothness*

Smooth chordwise and spanwise definitions of curvature, thickness distribution, and other properties are created using parametric cubic [57] and quartic [58,59] B-splines with few control points. The general B-spline parametric function is given as

$$S_i(t) = \sum_{r=0}^{n_t} P_{i+r} B_r(t); [0 \leq t \leq 1]$$

where, $P_i, P_{i+1}, P_{i+2}, P_{i+3}, P_{i+4}$... are the control points defining the B-spline, as shown in Figure 5 with $n_t$ as the B-spline order. The B-spline curve consists of many curve segments and is essential to match slopes of the curves to obtain a smooth continuous spline curve. It is also necessary to match the rate of change of slopes at joints since they are changing constantly.

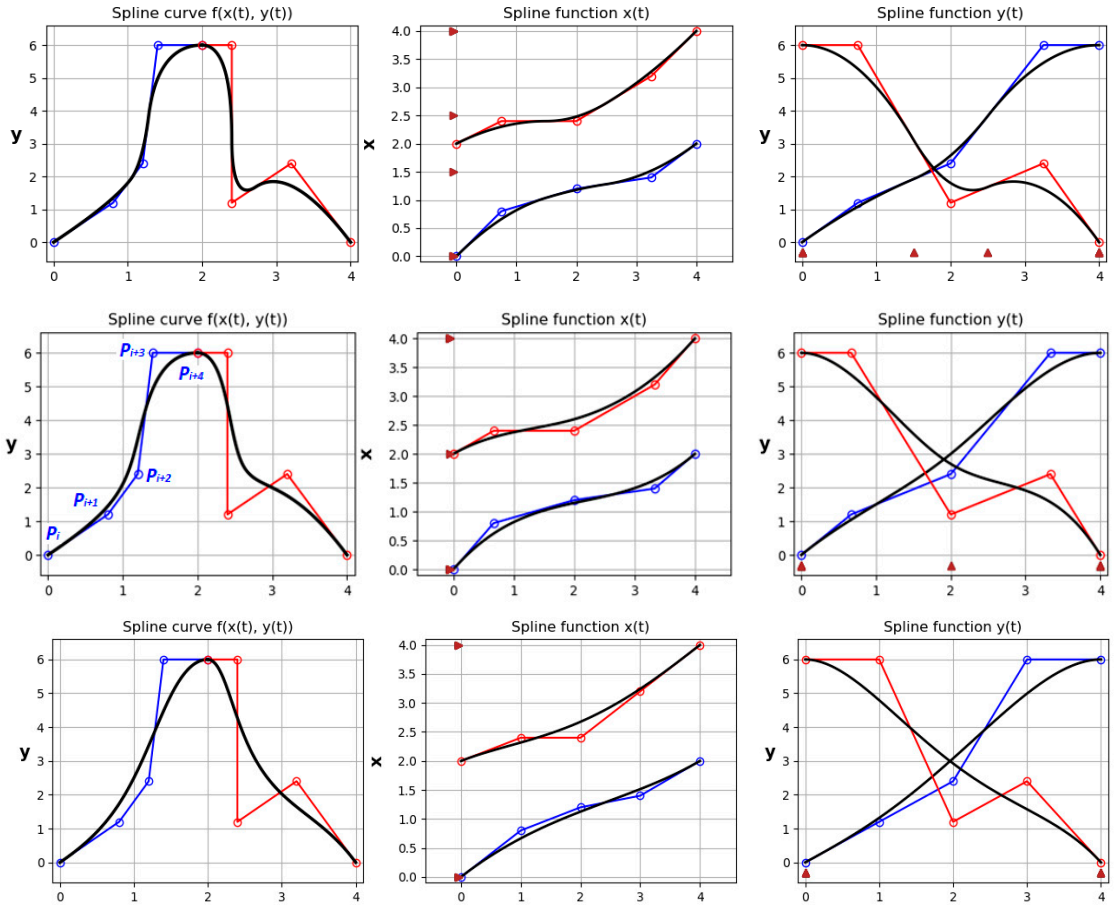



**Figure 5.** Parametric B-splines of degrees 2 to 4 row wise. All are C2 parametrically continuous.

Basis curves are defined such that first, second, and third levels of curve continuity are achieved parametrically as explained by Vince [57]. The first level of continuity, C0, ensures the physical end of one basis curve corresponds with the next one. The second level of continuity, C1, ensures the slope at the end of one basis curve matches that of the following curve. The third level of continuity, C2, ensures the rate of change of slope at the end of one basis curve matches the following curve. C continuity is harder to achieve than G continuity since it must satisfy magnitude and direction of the continuity [61]. Figure 5 shows continuity being satisfied in x(t) and y(t) curves for each degree of B-splines. One drawback of this method is that the spline curve does not pass through the two end control points. This property is necessary and is achieved as explained here for n control points. An extra control point $P_0$ is defined such that $P_1$ is at an equal distance from $P_0$

and $P_2$. Similarly, $P_{n+1}$ is defined such that, $P_n$ is at an equal distance from $P_{n+1}$ and $P_{n-1}$. This makes the curve segment pass through $P_1$ and $P_n$ (end control points) which now are the tangent points to the curve due to the phantom control points. The points $P_0$ and $P_{n+1}$ are not new input points but are defined internally using $P_1$ and $P_2$ and $P_{n-1}$ and $P_n$ as $P_0 = 2P_1 - P_2$ and $P_{n+1} = 2P_n - P_{n-1}$ relations. As an example, Figure 4 shows the chordwise curvature variation using cubic B-spline defined by five control points at a blade mid-section.

### *2.5. Streamwise Mapping and Stacking to Obtain 3D Airfoils*

Final steps in 2D blade section creation are stacking, rotation and scaling. The 2D stacking process is carried out in $(u, v)$ plane by translating each section to the user defined stacking position. Then rotation by stagger angle and scaling with chord value (*chrd*) are both performed to convert blade sections to the meridional plane. Stacking options are available as leading edge, percent chord, centroid, trailing edge, pressure side or suction side, and percent above or below meanline as shown in Figure 6. The offset value is subtracted from $(u, v)$ coordinates such that *u_stack*, *v_stack* coordinates align at the origin of $(u, v)$ system. An option of variable radial stacking spanwise is also available which can be utilized to control stacking at different spans. Once the stacking is performed for 2D blade sections, they are mapped correspondingly to the respective streamline coordinates to obtain the stacked $m'_{3D}$, $\theta_{3D}$ coordinates and x, y, z coordinates as shown in Figure 7. More details are provided by Siddappaji [48].

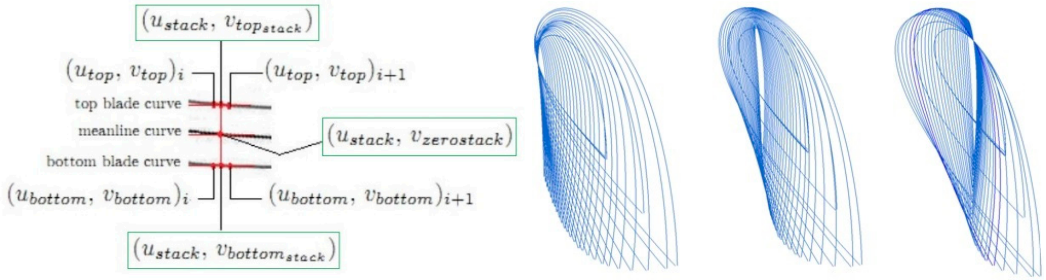



**Figure 6.** Coordinates on blade curves at a specified percent chord stack value with other stacking options. Wind turbine blade with 3D airfoils stacked on pressure side, meanline, and suction side from left to right.

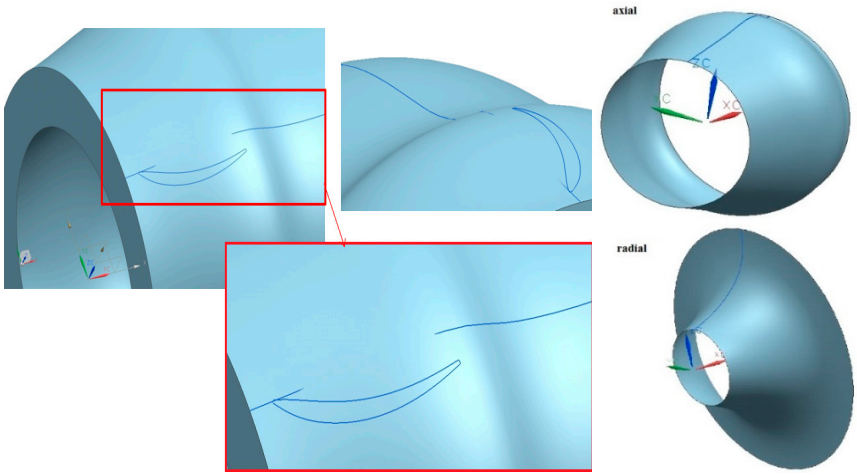



**Figure 7.** Mapping of the 3D airfoil on the corresponding stream surface (axial and radial).

Figure 6 shows a wind turbine with 3D airfoils stacked on pressure side, meanline, and suction side from left to right. Streamlines are defined using $x_s$ and $r_s$ coordinates—both upstream and downstream of the blade—for robust mapping. Defined number of

streamlines pass through a blade geometry. 2D blade sections created are mapped with their corresponding stream surfaces (surfaces of revolution of 2D streamline curves) to obtain the cylindrical coordinates transformed into Cartesian coordinates to create 3D blade shapes.

### *2.6. Parameters for Geometry Generation in 2D and 3D*

Several parameters are defined for the blade geometry construction. An entire list of parameters is given in Figure 8, which completely opens the design space. Spanwise definition of chordwise curvature and thickness through a few B-spline control points is convenient in creating smooth blades and this feature is also extended to leading and trailing edge thickness as demonstrated in Figure 9 for the HPC rotor.

| | | |
|---|---|---|
| casename | | legend: |
| N_blades | | dimensional |
| bsf (scaling factor) | | non-dimensional |
| N_streamlines | | |
| camber (meanline) | *uv 2D airfoil import* | |
| | *curvature* | *chordwise spline* |
| | *curvature* | *chordwise+spanwise spline* |
| thickness | *wennerstrom* | |
| | *quartic spline* | |
| | *quartic spline+sharp TE* | |
| | *modified NACA 4 digit* | |
| thickness multiplier | *spanwise spline* | |
| LE | *straight line* | |
| | *spline* | droop |
| TE | *straight line* | |
| | *spline* | |
| chord | *actual as input* | |
| | *spline multiplier* | |
| Metal Angles at IN and OUT | *table definition at each streamline* | |
| | *IN-OUT angles spanwise spline* | |
| | *incidence and deviation spanwise spline* | |
| lean and sweep | *tangetial lean and axial sweep* | *chordwise spline* |
| | *tangetial lean and axial sweep* | *chordwise+spanwise spline* |
| | *true lean and true sweep* | *chordwise spline* |
| | *true lean and true sweep* | *chordwise+spanwise spline* |
| parameters at each streamline | *airfoil type* | |
| | *u_stack, v_stack %* | |
| | max. thickness to chord ratio (tm/c) | |
| | LE thk | |
| | TE thk | |
| stacking | *LE* | |
| | *TE* | |
| | *%chord* | |
| | *%meanline (above or below) +%chord* | |
| | *centroid* | |
| | *variable spanwise* | |
| spanwise B-SPLINE varation | *sweep* | axial |
| using n control points | *sweep* | true |
| | *lean* | axial |
| | *lean* | true |
| | *metal angles* | |
| | *stagger angles* | |
| | *incidence and deviation* | |
| | *chord* | |
| | *tm/c* | |
| hub offset | | |
| tip offset | | |
| streamlines x-r | *normalized by LERtip* | |

**Figure 8.** Non-dimensional and dimensional parameters for geometry generation.

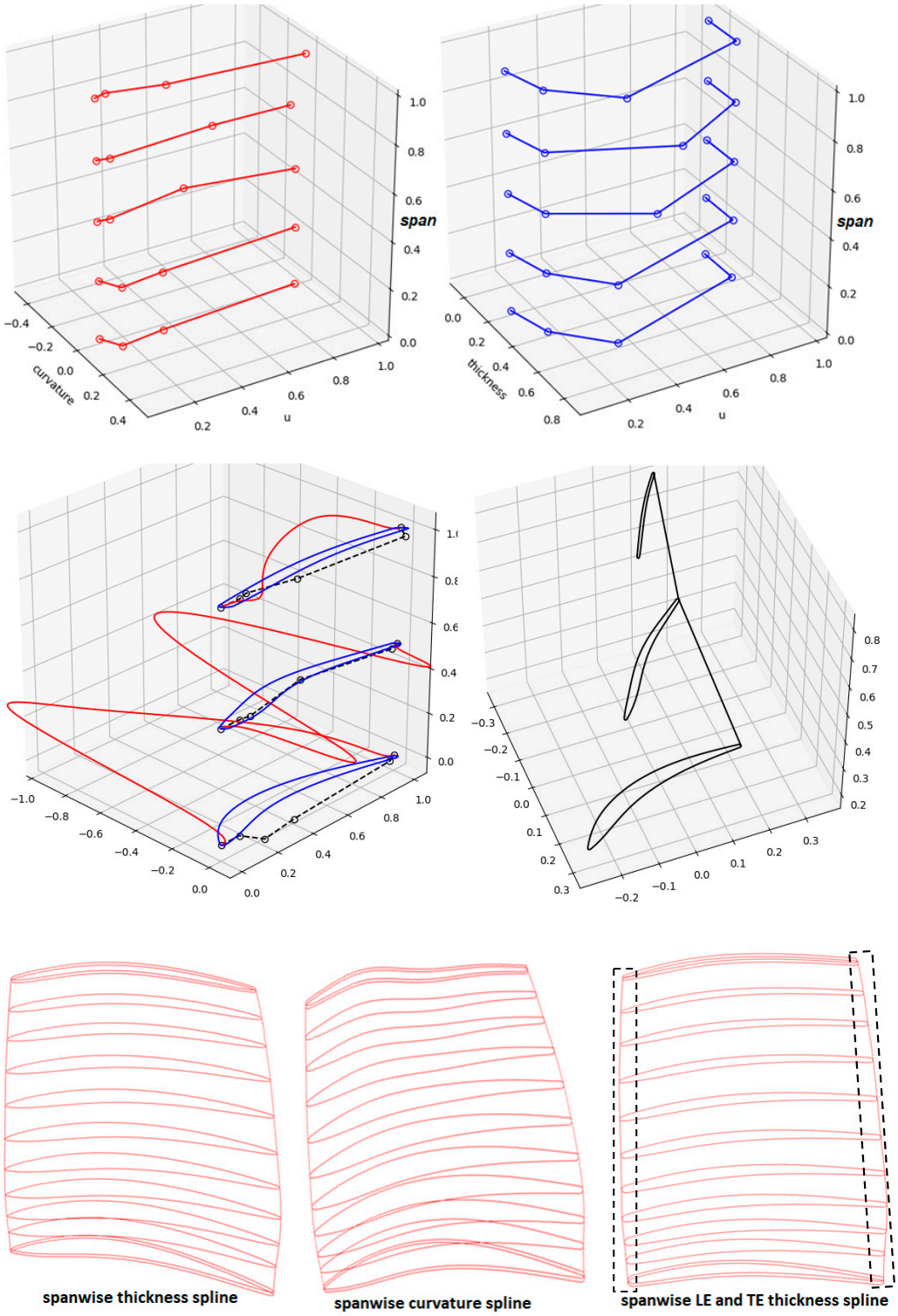



**Figure 9.** Spanwise and normalized unstaggered chordwise (u) B-spline control points of curvature (left) and thickness definition. 2D staggered airfoils constructed with the right chord definition are also shown. The last row shows spanwise spline for curvature, blade, and LE and TE thickness.

Most inputs for blade generation are non-dimensional. It is obtained by normalizing the coordinates by leading edge tip radius. Blade scaling factor is used to generate dimensional 3D airfoil. Several angles are defined as inputs along with incidence and deviation varied spanwise. Purely axial, purely radial, axial inlet–radial outlet, and radial inlet–axial outlet are the four options available which cover all types of turbomachinery.

### *2.7. Blade Visualizers*

Using 2D and 3D matplotlib libraries of Python language, several geometry visualizers are created for the benefit of designers. Meridional view, 2D blade cascade view, 3D blade view, and 3D rotor–stator view are crucial in understanding the design as shown in Figure 10 for several axial turbomachines. Curvature of the blade section can also be visualized to examine smoothness of surfaces, LE, and TE as shown.

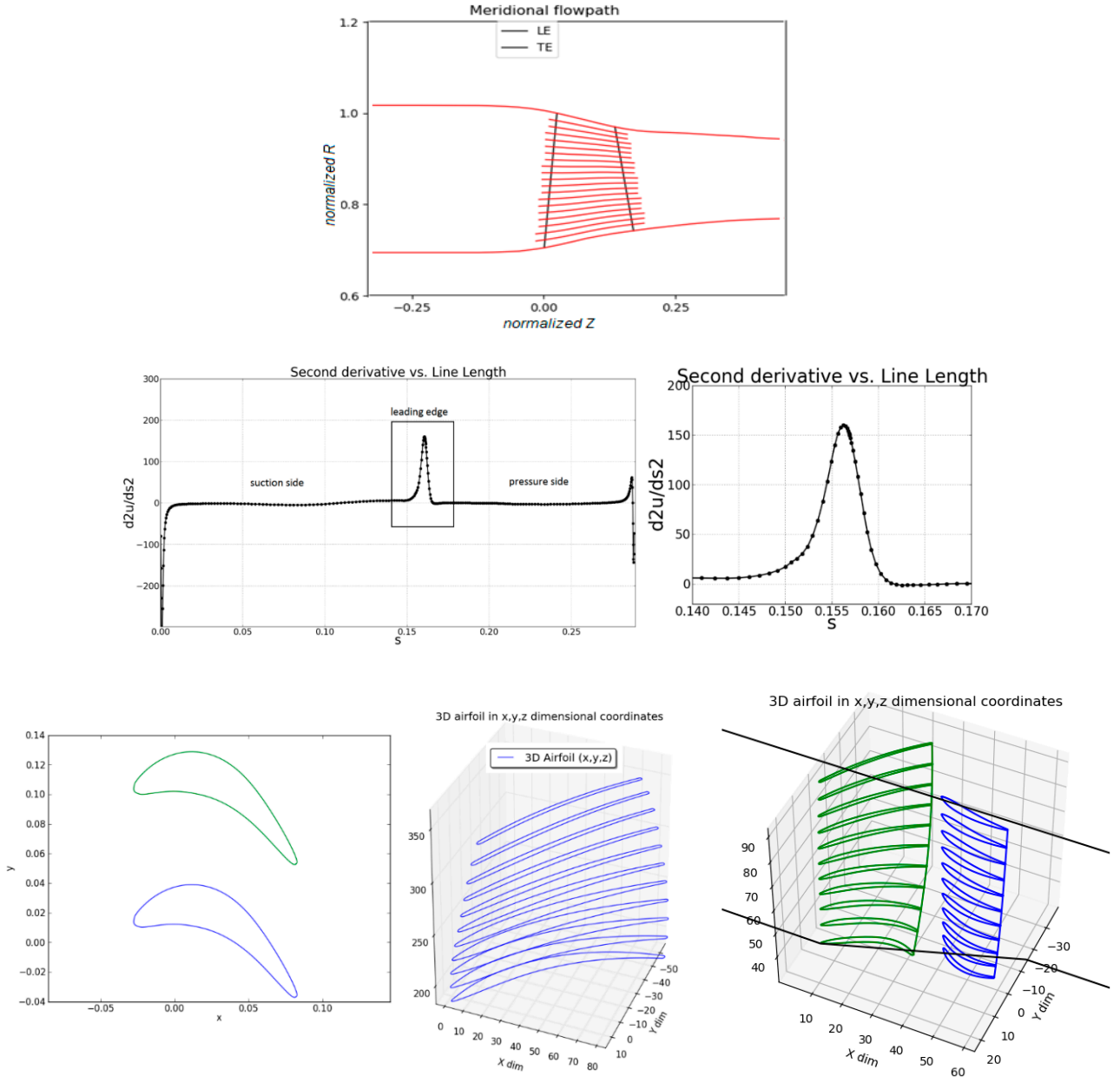



**Figure 10.** Meridional view of transonic compressor rotor, airfoil curvature, 2D cascade of an axial turbine rotor, 3D blade view of a compressor, and a sample turbofan stage 3D view as examples of blade visualizer in use.

### *2.8. Periodic Boundary for Fluid Domain and 2D Grids*

The geometry generator can also create periodic boundaries as an option to create top and bottom periodic domains with upstream and downstream domains useful in creating fluid domains for 3D CFD analysis. It generates a 3D meanline curve and offsetting it above and below by half the pitch, a periodic boundary is created as shown in Figure 11. The

upstream and downstream domains are created by extending the meanline with respective slopes to maintain curve continuity. 2D grids for the blades as O grid and H grid for the background domain can also be created as seen in Figure 11. An option for elliptic smoothing of the grids [62] is also embedded in the tool. Latest version of the tool has isolated this capability to develop into a stand-alone research effort for 3D grids.

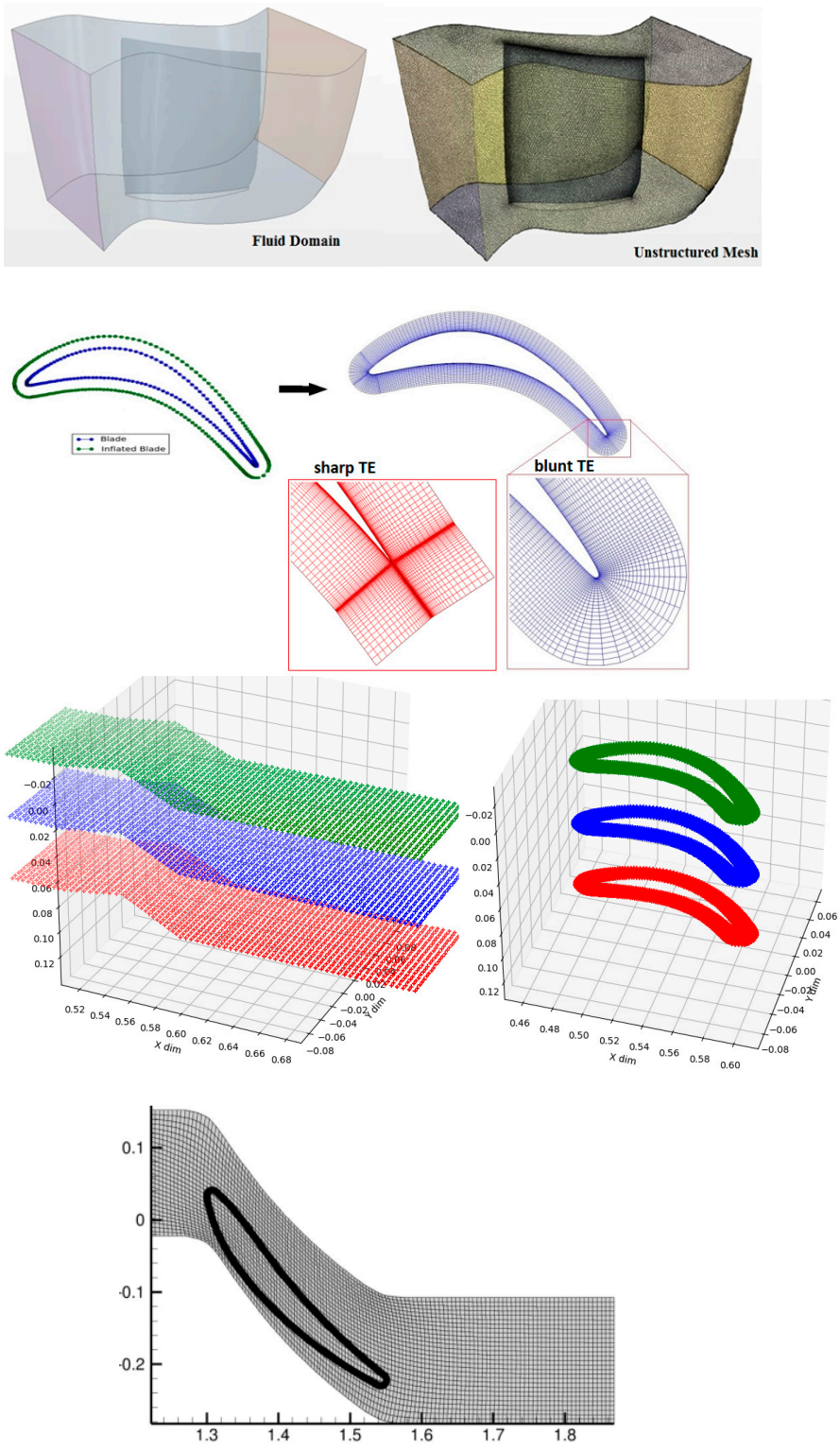



**Figure 11.** Periodic boundary curves created and imported in a CFD solver for meshing. 2D grids created for round and sharp trailing edge with an elliptic smoother option for overset 2D mesh.

### *2.9. Splitter Blade, Sweep, and Lean Effect*

Blade offsets in tangential direction to create splitters and non-axisymmetric blades to accommodate boundary layer ingestion can be easily created by defining a tangential lean or true lean as a function of blade pitch. A transonic fan with a splitter consisting of 24 rotor blades oriented in periodic order with 12 main blades and 12 splitters was used to create a baseline for a multi-objective multifidelity design optimization using the geometry generator [63]. Splitter blades are created by defining constant spanwise tangential lean of half the pitch for the entire blade and different angles, leading and trailing edge, than the main rotor as shown in Figure 12. Splitters can easily be defined in between full blades and the tool provides freedom to place them at any tangential and meridional location in the passage between blades. Another application which utilized a splitter rotor for a high-pressure turbine with overset grids and analyzed for Euler solution using a high-order in-house 2D Discontinuous Galerkin Harmonic Balance solver [64,65] at low Reynolds number to show the splitter effect on velocity can be seen in Figure 12.

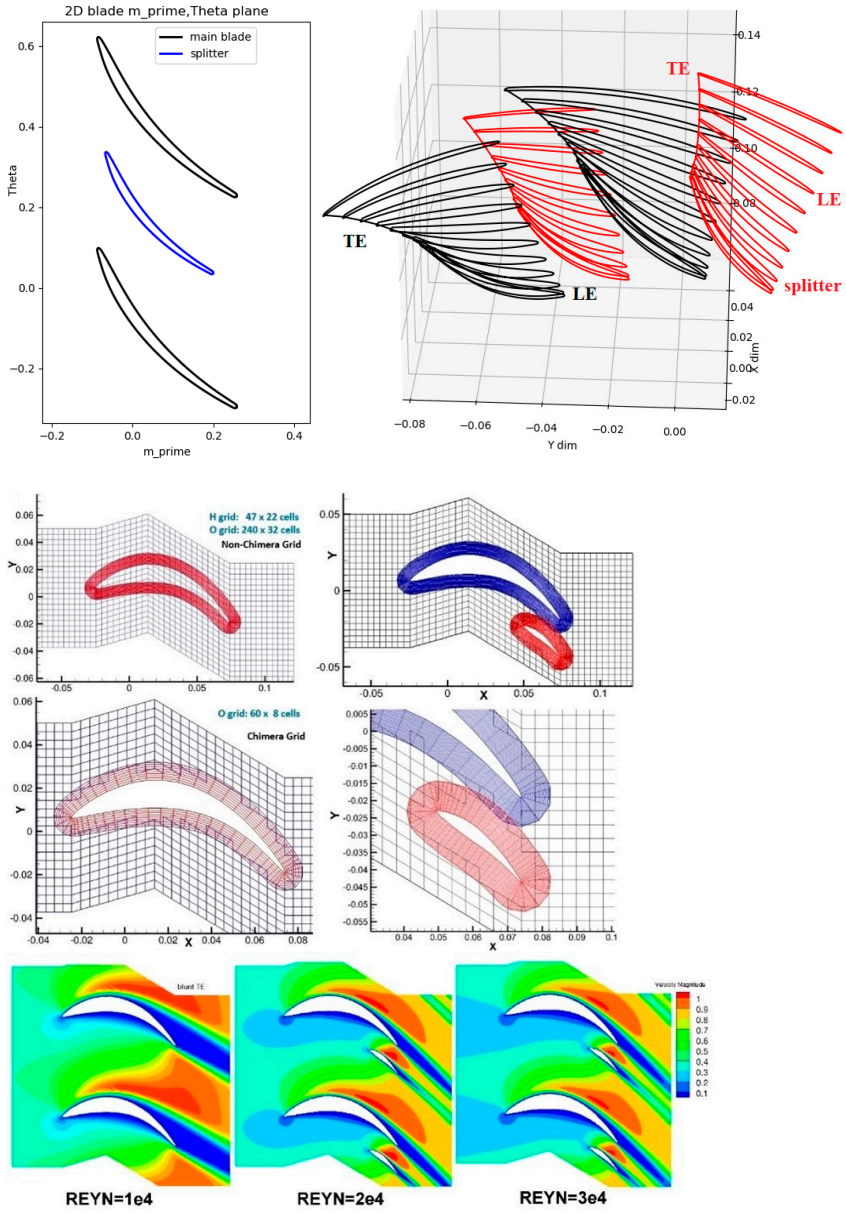



**Figure 12.** Blade to blade mesh at 50% span of the transonic rotor with the splitter. Chimera grid of the turbine rotor and splitter rotor solved in a higher order in-house DGHB solver.

Spanwise sweep and lean can be defined in two ways, and the effect on efficiency is shown by Neshat et al. [66]. True sweep is along the chord and true lean is normal to the chord, whereas meridional sweep is in the meridional direction and tangential lean is in the tangential direction as shown in Figure 13. The geometry generator has both options and they are defined by a cubic B-spline using a specified number of control points spanwise. The corresponding $\delta m'$ values at each streamline for sweep are evaluated by interpolating the $\delta m'$ curve at each spanwise location on each streamline as depicted in Figure 13. Similarly, lean can be added with few control points locally to eliminate corner separation and improve adiabatic efficiency of an isolated rotor.

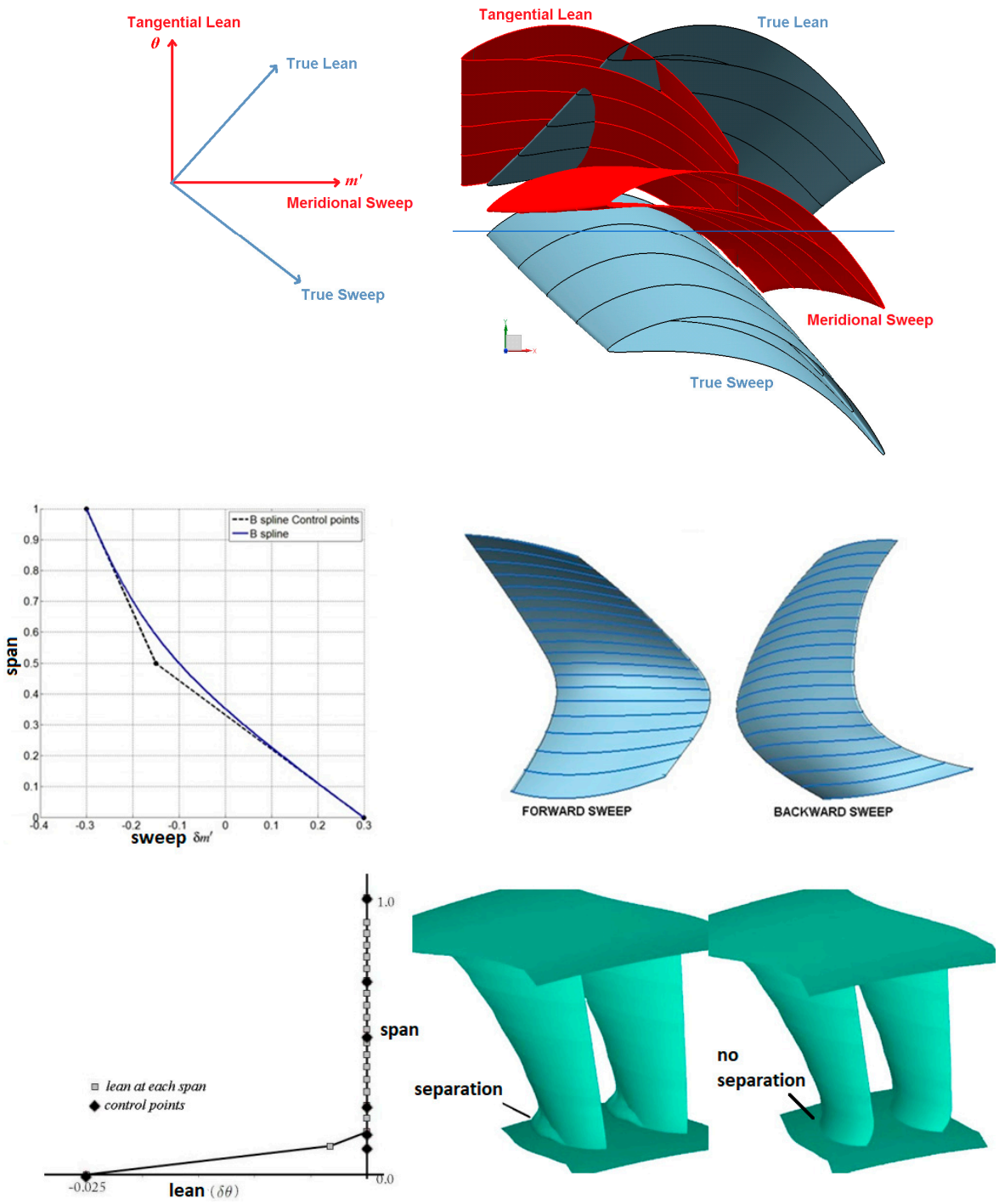



**Figure 13.** Two types of sweep and lean with spanwise meridional sweep and local lean.

### *2.10. Automation, Optimization, and CAD Connection*

The tool outputs a 3D blade in a specified number of data files containing defined number of coordinates of 3D blade sections which can be imported into a CAD package to obtain a smooth lofted blade. Parametric definition of splines makes the geometry modification process quicker and easier, and it allows exploration of some novel concepts. All Boolean operations are carried out in CAD to obtain a watertight solid. It can easily be tied to any automation and optimization chain. The free open-source DAKOTA program [67] from the Sandia National Laboratories is used for optimization. DAKOTA (Design Analysis Kit for Optimization and Terascale Applications) provides a flexible, extensible interface between analysis codes and iterative system analysis methods. It works in parallel with several optimization options and runs under Linux, Unix, and Linux-like environments (Windows with Cygwin). Figure A2 in Appendix A shows the general flowchart for the optimization process and geometry generator connected to a CFD solver and FEA tool.

### *2.11. 3D Section Offset and Tool Generality*

In some cases, grid generators require an extruded blade due to tolerance issues and they are useful for other purposes as well. It is achieved through a simple offset of hub and tip streamline coordinates in the normal direction to streamline. At any point, $m'_s$ normal in x-direction, $x_{NORM}$ and normal in r-direction, and $r_{NORM}$ are calculated by using orthogonal property between normal and slope at that point. Newly extruded hub and tip streamline coordinates are used instead of original coordinates with blade parameters corresponding to the original hub and tip streamlines as input to obtain an extruded 3D blade as shown in Figure 14. The offset in normal direction, $a$, is a percent offset and is multiplied by reference length (tip radius of LE) to obtain $\Delta n$ which is used to calculate $b$, an intermediate extruded blade scale factor. Newly extruded coordinates for offset hub and tip streamlines are calculated using normal and $b$ from the following equations:

$$x_{NORM} = \frac{dr_s}{dm'_s} \; ; \; r_{NORM} = -\frac{dx_s}{dm'_s}$$

$$\Delta n = aL_{ref} \; ; \; \Delta n = b\sqrt{{x_{NORM}}^2 + {r_{NORM}}^2}$$

$$x_{s_offset} = x_{sHUB} + bx_{NORM} \; ; \; r_{s_offset} = r_{sHUB} + br_{NORM}$$

$$x_{s_offset} = x_{sTIP} - bx_{NORM} \; ; \; r_{s_offset} = r_{sTIP} - br_{NORM}$$

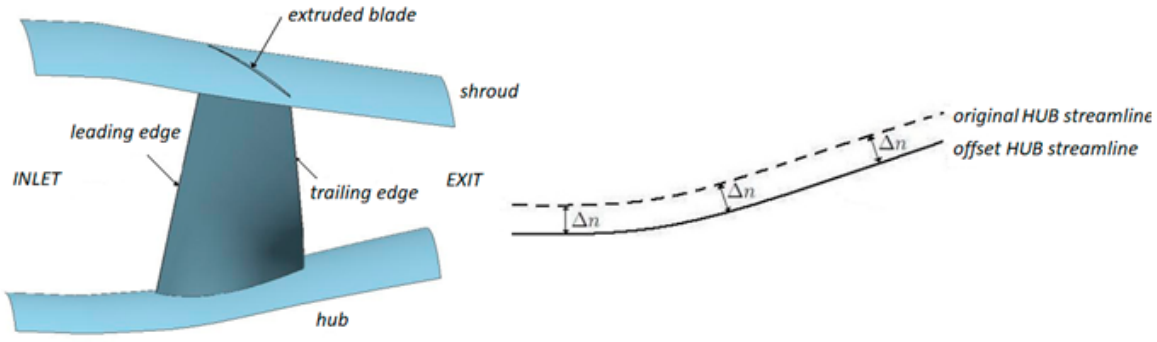



**Figure 14.** Normal offset of hub streamline and extruded blade at the tip.

Options to define blade angles for axial and radial leading and trailing edges, incidence, deviation, stagger, and spanwise splines to control these values makes the tool very general and it can handle a wide variety of turbomachinery as shown in Figure 15. Spanwise spline definition of chord, thickness, sweep, and lean of the blades are beneficial to create traditional and non-traditional shapes which are smooth due to C2 continuous cubic or quartic B-splines.

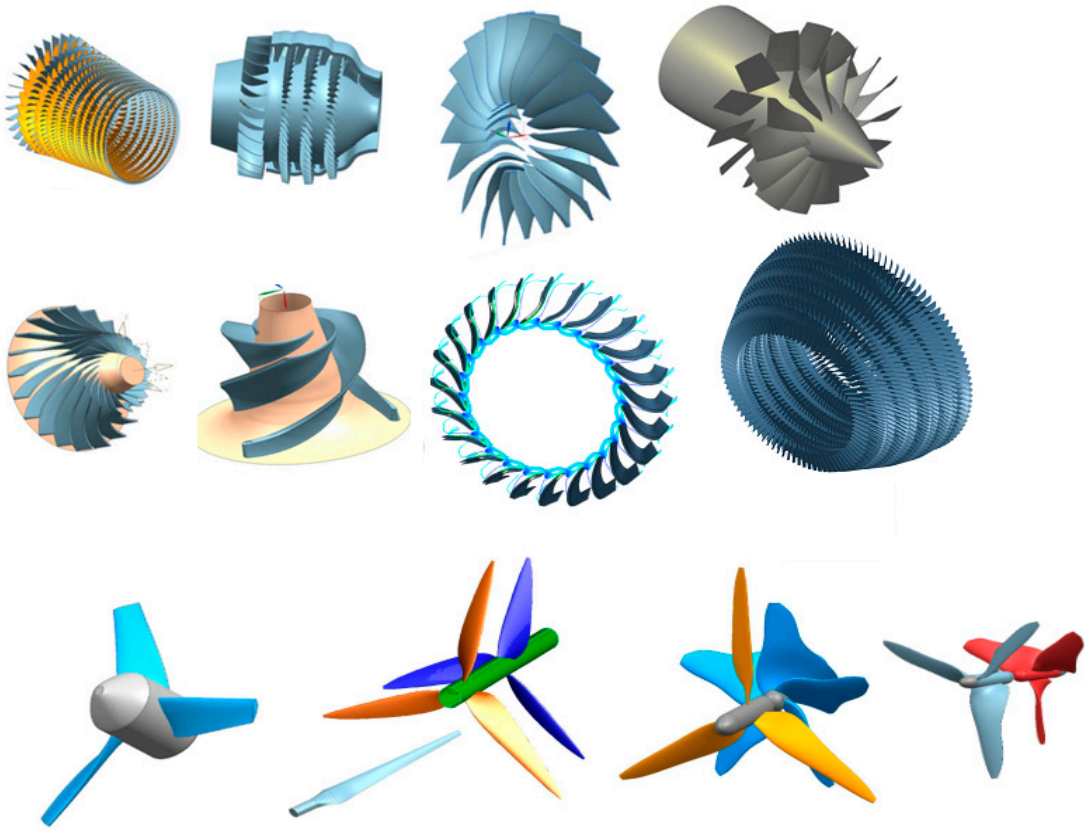


**Figure 15.** General capability of the tool showing ducted and unducted rotors.

### *2.12. Parametric Sensitivities*

Calculating sensitivities of design parameters is very important to distinguish between significant and non-significant parameters. This knowledge is essential in minimizing the optimization cycle using a cost-efficient set of parameters. Fundamentally, calculating derivatives of a function is the basis for sensitivity analysis. This can be performed analytically, algorithmically, or numerically using finite difference schemes. A sensitivity calculator based on accurate and fast finite difference scheme [29,30] is connected to the geometry generator and several design parameters are enabled to create perturbations and perform sensitivity analysis. The normal component of the computed sensitivity is taken as the sensitivity of a point on a face driven by a design parameter.

Figure 16 shows three different perturbations of TE blade angles for a representative sixth stage rotor of 10 stage $E^3$ HPC. $\frac{d(norm)}{d(out_beta)}$ is the sensitivity plotted for each point on the blade surface when the parameter *out_beta* is changed and shows that it can be positive or negative depending on the direction of geometrical perturbation. The second image in Figure 16 shows the sensitivity to all TE blade angles changing with blue as negative direction and red as positive direction compared to unperturbed geometry. The third image shows that the sensitivity to the TE blade angle changed only at the hub section.

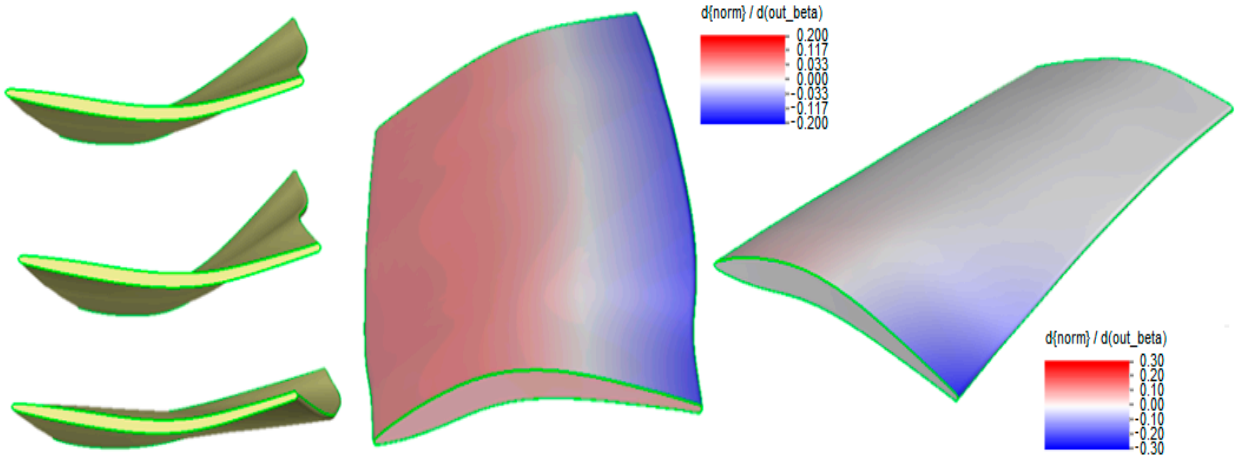
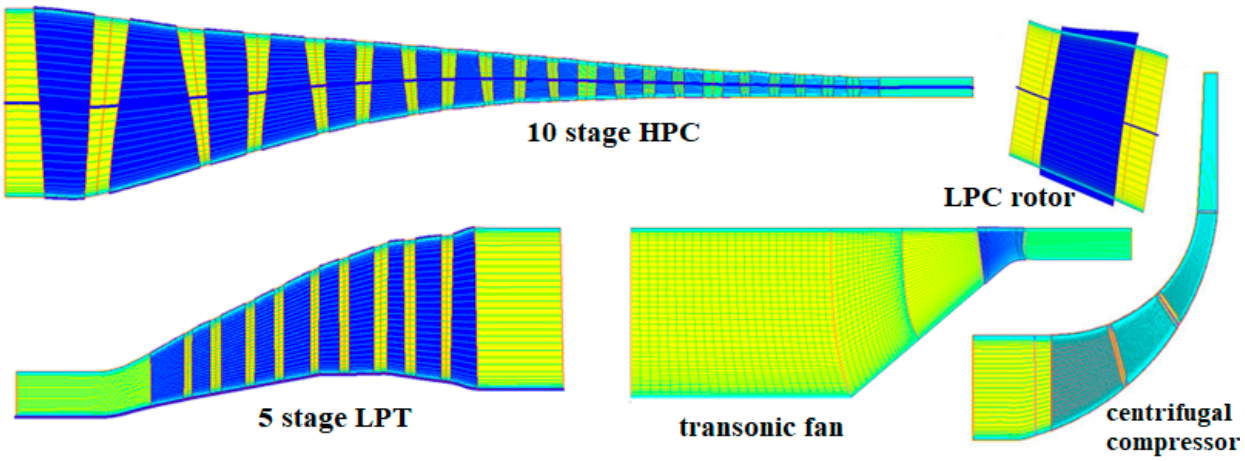
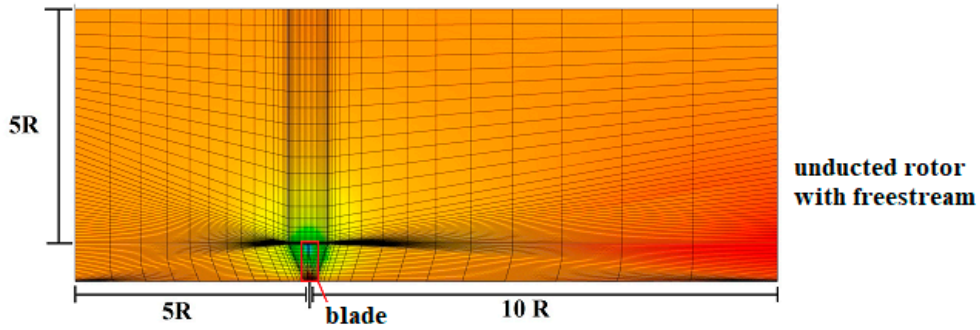



**Figure 16.** TE perturbation with three different spanwise distributions (**left**); sensitivity to all TE blade angles changing (**center**) and to TE blade angle at hub alone.

### *2.13. 3D CFD Simulation Setup and Grid Dependency Study*

3D geometry created is analyzed in RANS solvers to estimate aerodynamic performance of these shapes. The entire framework of multifidelity multidisciplinary design analysis and optimization is shown in Appendix A. Numeca's Fine/Turbo RANS solver [68] with Spalart–Allmaras turbulence model is used for majority of cases and for few, StarCCM+ RANS solver [69] is also used. Wherever applicable, an AGS transition model is used. Inlet boundary condition of total pressure, temperature, and default turbulent viscosity normal to the inlet face are applied for ducted machines and static pressure or mass flow at exit face. Grid dependency studies are appropriately carried out for each configuration and solver to obtain the best grid metrics which are described here categorically for several examples [40,70,71]. Figure 17 represents various solo blade or stage domains created for devices such as $E^3$ 10 stage HPC; $E^3$ 6 stage LPT; solo LPC rotor; transonic fan rotor with splitter; radial compressor novel configuration; and unducted rotors including propellers, wind, and hydrokinetic turbines among others. Grid dependency for each category of machines is shown with a few examples in Figure 17.




| Type | Device | Example | Solver | Grid points | ETA |
|---|---|---|---|---|---|
| **ducted** | axial | *transonic fan* | Fine/Turbo | 52,360 | 0.775 |
| | | | | 364,194 | 0.784 |
| | | | | 2,706,658 | 0.787 |
| **ducted** | axial | HPC rotor | StarCCM+ | 1,719,696 | 0.912 |
| | | | | 10,049,545 | 0.921 |
| | | | 2X prism layers | 10,226,494 | 0.920 |
| **ducted** | radial | impeller | Fine/Turbo | 38,544 | 0.776 |
| | | | | 1,882,816 | 0.828 |
| | | | | 2,851,120 | 0.825 |
| **unducted** | | *hydrokinetic turbine* | StarCCM+ fullfrustum 16 mm | 403,434 | 0.111 |
| | | | full frustum 6 mm | 5,385,607 | 0.591 |
| | | | full frustum 5 mm | 7,018,797 | 0.574 |
| | | | | | **Thrust [N]** |
| **unducted** | | propeller | Fine/Turbo | 751,356 | 878 |
| | | | | 1,124,486 | 847 |
| | | | | 1,408,264 | 849 |
| **unducted** | | wind turbine | Fine/Turbo | 880,307 | 1209 |
| | | | | 1,366,195 | 1194 |
| | | | | 1,427,515 | 1195 |

**Figure 17.** Axisymmetric views of 3D domains for ducted axial 10-stage HPC, 5-stage LPT, solo LPC rotor, centrifugal compressor novel configuration, and unducted rotors without nose at hub. Grid dependency for listed examples of configurations are also listed in the bottom image.

## 3. Results

A wide variety of smooth geometries have been created using the described parametric approach at 2D and 3D levels by varying only a few parameters defined by B-splines or otherwise as demonstrated, targeting reduced flow separation, improved diffusion, and radial flow with entropy minimization.

### *3.1. Ducted Axial Turbomachines*

Axial compressors, turbines, and turbofan designs are described in this sub-section.

#### 3.1.1. First 6 Stages of 10 Stage $E^3$ HPC

$E^3$ 10 stage high pressure compressor consists of 21 blade rows including an inlet guide vane [72]. Geometry generator is used to create the blade rows with spanwise variable angular momentum definition obtained from the report and through T-Axi axisymmetric solution with non-straight leading and trailing edges. Blade metal angles and thickness-to-chord ratio data are interpolated from the report. The first five rotors are stacked at the area centroid of each blade section and the rest at quarter chord. Figure 18 shows metal angles comparison of rotor 3 between the design tool and NASA report, demonstrating the tool's accuracy. Figure 18 shows 3D geometry of all 21 blade rows which was used by NASA to define the $E^3$ geometry for a complete preliminary engine simulation [73] and as a baseline for further analysis. The first six stages were analyzed in an unsteady Numeca solver as part of a class project by Matthys et al. [74] in collaboration with authors using non-linear harmonic method which converts time into frequency domain to speed up solution time. Figure 19 shows entropy at 50% span at a single point in time. Wake interaction can be seen distinctly and the associated entropy as loss near trailing edges.

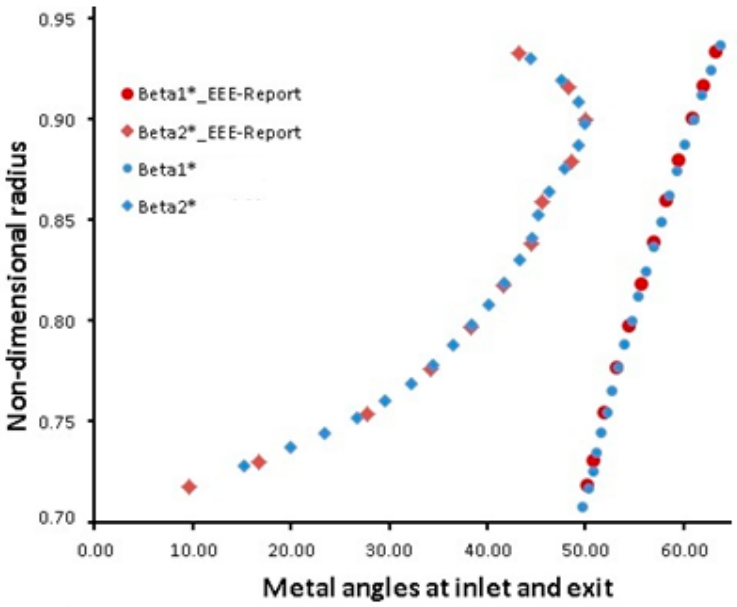



**Figure 18.** *Cont.*

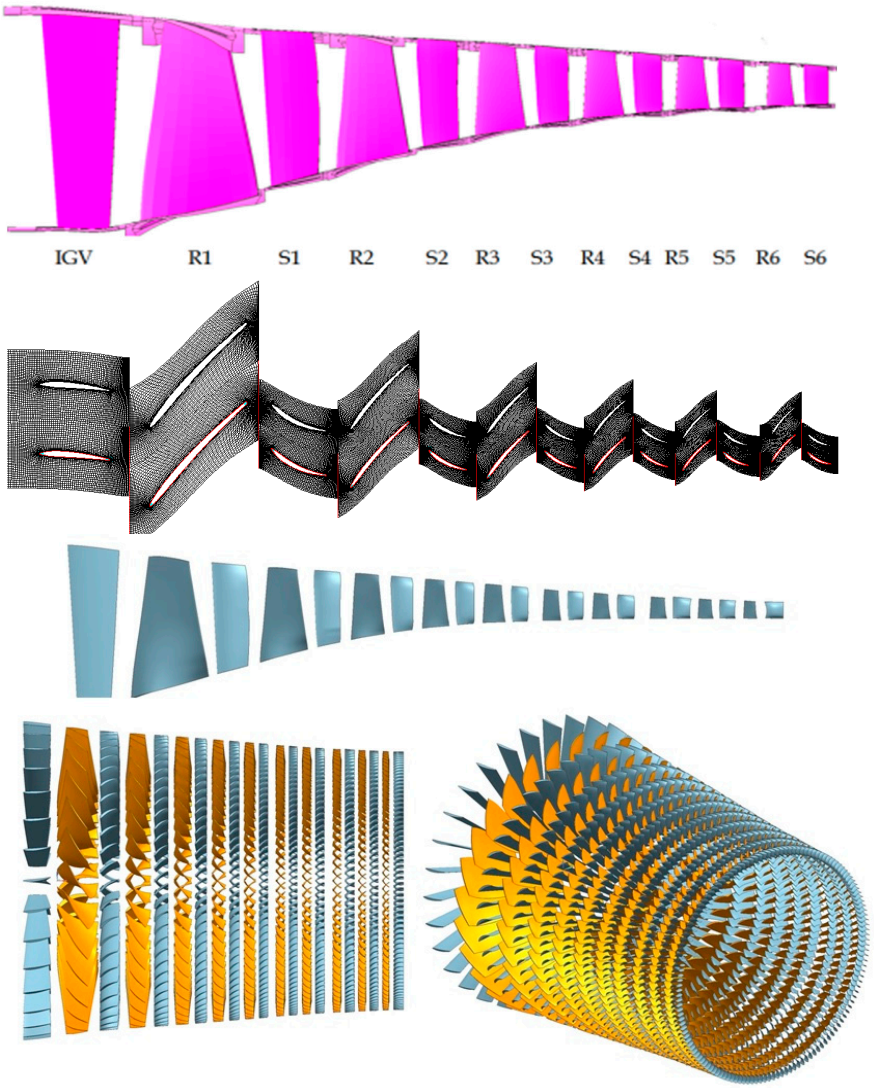



**Figure 18.** Comparison of blade angles of $E^3$ HPC rotor 3 generated by the tool with NASA report along with grid representation of first 6 stages and CAD for all 10 stages.

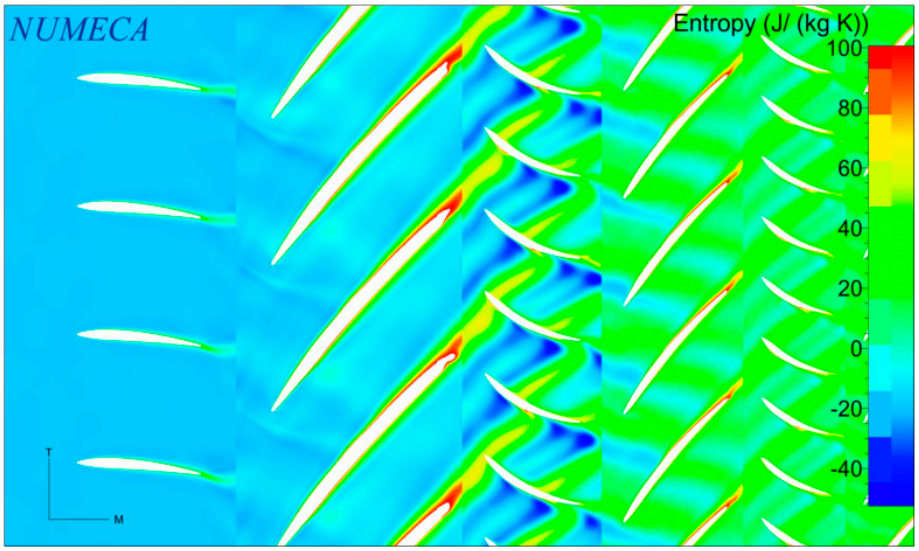



**Figure 19.** Entropy at a single point in time from an unsteady solution at 50% span [74].

### 3.1.2. $E^3$ 5 Stage LPT

The five-stage LPT of the $E^3$ representative configuration [75] is used as an example to demonstrate the effect of streamline curvature in blade calculation. When the streamline slope, $\varphi_{mz}$, is non-negligible, axial and meridional velocities are different and this must be considered in meanline design. This difference in velocities affects the blade count and blade angles to be used in generating 3D blade shapes as shown in Figure 20. An in-house 1D meanline tool for an axial turbine was used to generate these values.

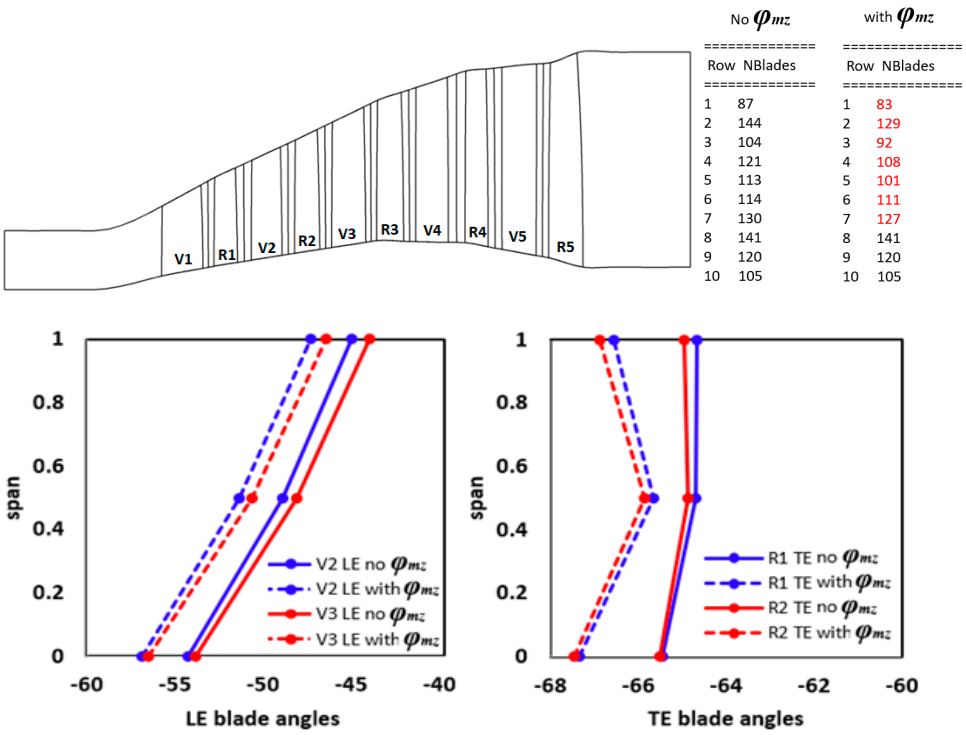



**Figure 20.** Five-stage $E^3$ LPT representative configuration showing different blade count and blade angles due to streamline slope being accounted for in a meanline axial turbine design solver.

### 3.1.3. Split Tip $E^3$ HPC Rotor 6

Utilizing spanwise local lean capability, a novel split tip blade geometry is created using the sixth rotor of a highly loaded 10-stage $E^3$ HPC representative configuration [72]. Two identical blades are leaned tangentially in equal amount and in opposite directions from 85% span onwards as shown in Figure 21. The blades are cut spanwise at mid-chord and then merged to obtain the split tip similar to alula feathers on a soaring bird as described by Srinivas [76] in collaboration with the authors.

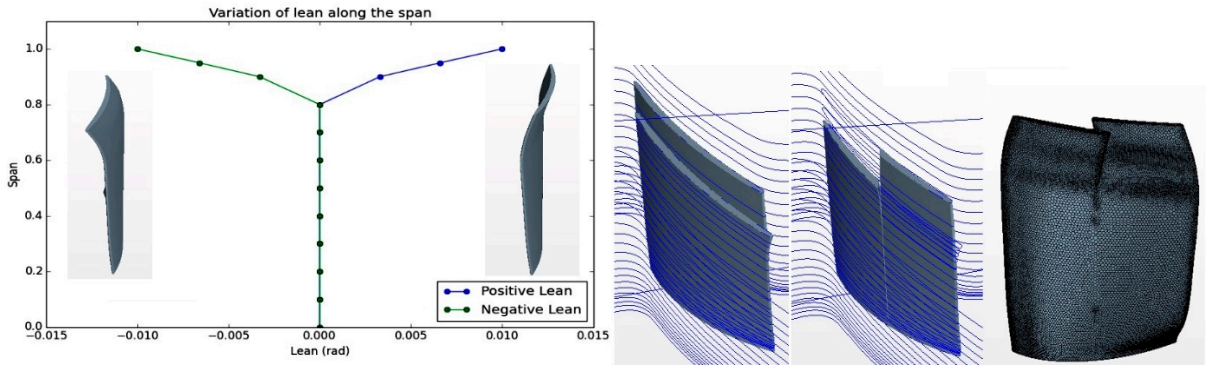



**Figure 21.** Local lean defined in both directions near the tip and merged to create a split tip.

Figure 22 shows the peak efficiency for both rotors at various tip clearances and is evident that the novel rotor performs better at higher clearances. It also shows the streamtubes of entropy for both rotors, and flow mixing is visible at the split. Relative mach contours at 90% span of the two rotors at stall, operating point, and choke are shown for tip clearances of 5.00% in Figure 23.

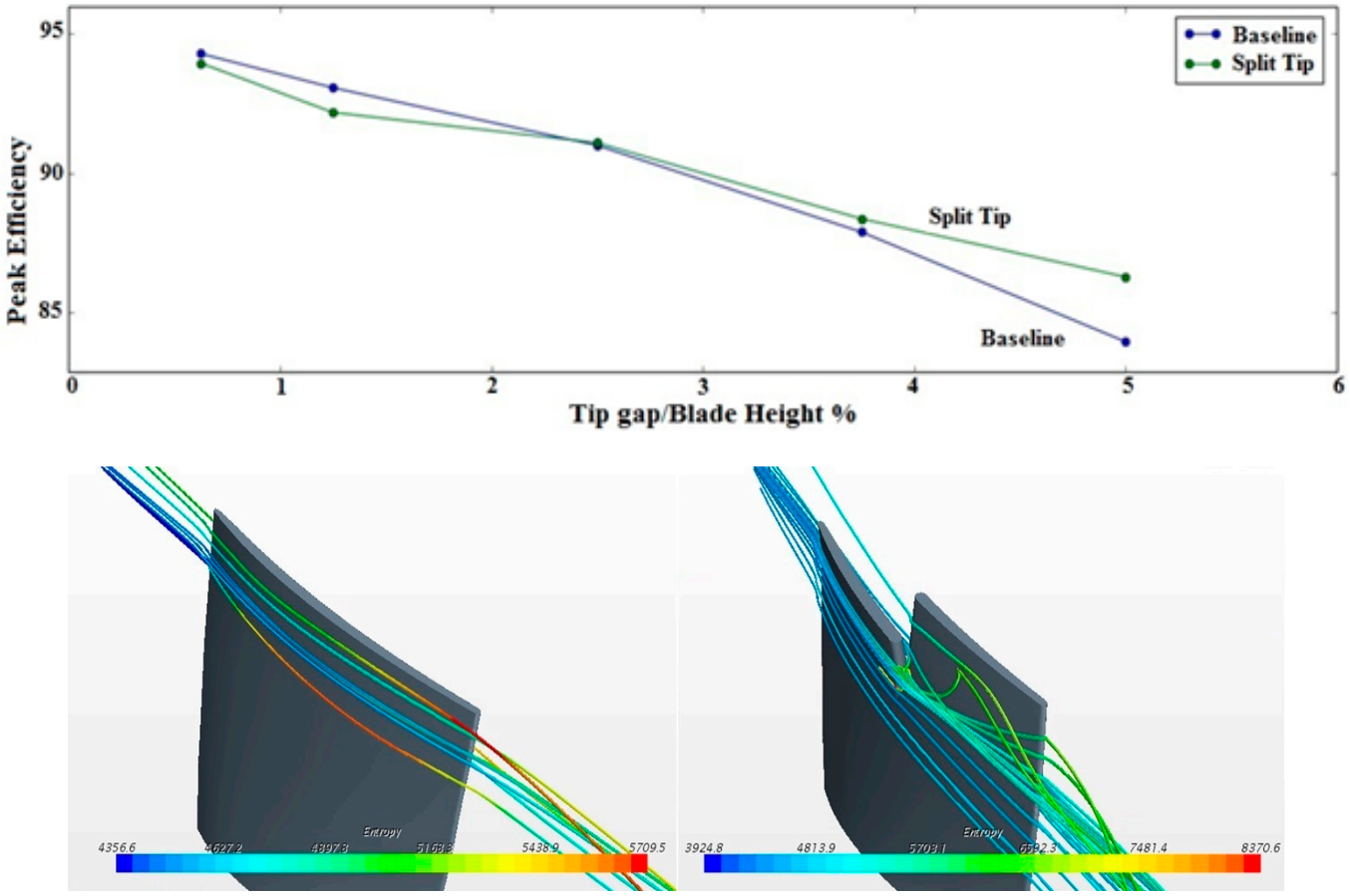



**Figure 22.** Streamtubes of entropy for baseline and split tip case.

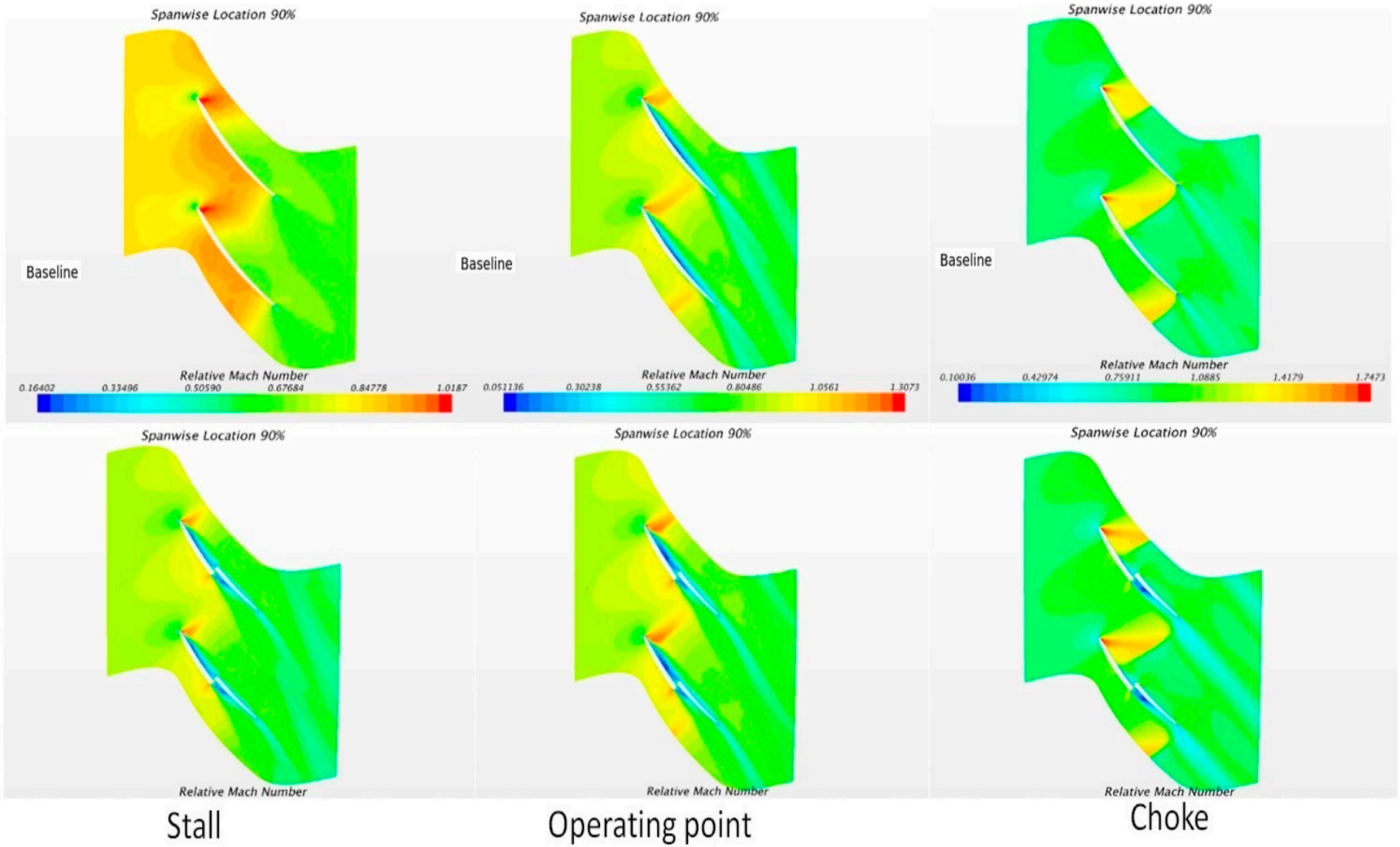



**Figure 23.** Mrel at 90% span for both rotors at stall, operating point, and choke at 5% tip clearance.

#### 3.1.4. Non-Axisymmetric OGV for Boundary Layer Ingestion Fan

A non-axisymmetric outlet guide vane is created for a fan stage with inlet distortion. Spanwise lean perturbation and in the circumferential direction is introduced asymmetrically to take advantage of flow non-uniformity originating at the fan inlet installed on the rear of an aircraft rather than under the wing, as described by Kumar and Mandal [77–79] in collaboration with the authors. Figure 24 shows axisymmetric and non-axisymmetric OGV.

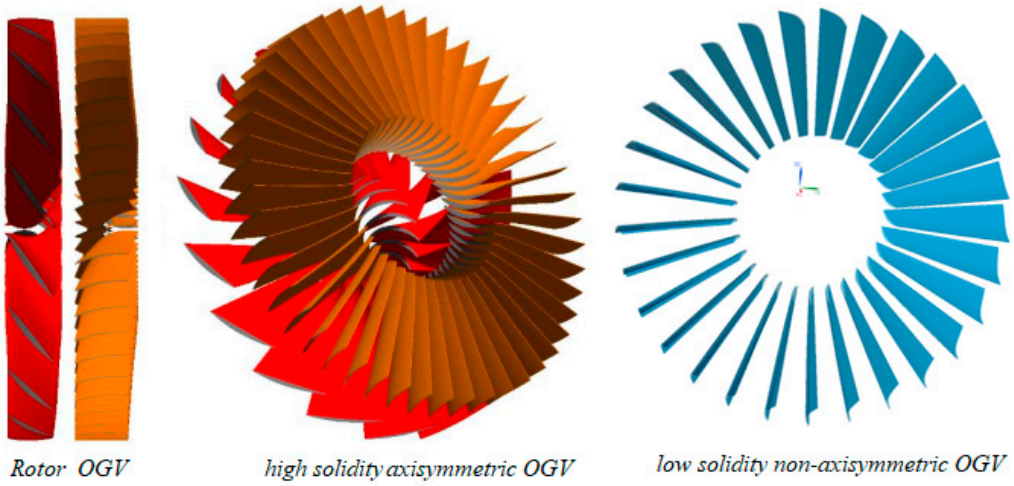



**Figure 24.** Non-axisymmetric OGV defined using lean for a boundary layer ingestion fan.

### *3.2. Ducted Radial Turbomachines*

This sub-section demonstrates the capability of designing radial turbomachines. Several options in the input parameters for inlet and exit angles to be defined exist which include purely radial, axial inlet with radial exit, and radial inlet with axial exit. Another check embedded in the tool to distinguish between axial and radial machines is by evaluating the slope of streamline and for radial machines, $\left|\frac{dr_s}{dm'_s}\right| > \left|\frac{dx_s}{dm'_s}\right|$ should hold true. When this condition is satisfied, normalized $r_s$ values (in radial cases) are used to obtain $m'_{TE}$ (or $m'_{LE}$ when the flowpath starts as radial) through inverse spline, instead of the usual method of using normalized $x_s$ values (in axial cases) while calculating the non-dimensional meridional chord. The 3D blade sections are created and stacked with the trailing edge as the stacking axis.

#### 3.2.1. Centrifugal Compressors

A blade shape similar to the low-speed centrifugal compressor [80] was created to demonstrate the generality of the design tool. The NASA report [81] contains hub and tip streamline coordinates and other airfoil geometry data and is also used by Sato et al. [82] for investigating the rotor–stator interaction and impeller performance. The inlet axial angle is taken as zero and the outlet radial angle as 90 degrees. Figure 25 shows the centrifugal compressor initial and optimized angle design. The streamwise coordinates required for blade geometry generation are taken directly from the report. The camber line of the airfoil section is not defined for the model in the report and hence an optimization with respect to the curvature of the camber line is performed using DAKOTA optimizer. The camber line of the optimum blade obtained is S-shaped as demonstrated by Mishra [83] in collaboration with the authors. The baseline was designed with 82% efficiency for pressure ratio 1.17. The percentage gain in efficiency and pressure ratio is 8.55% and 2.97% respectively. Figure 25 shows control points and cubic B-spline defining chordwise variation in curvature for mid-section of baseline. Figure 26 shows contours of entropy and relative mach number at 95% span for the S-shaped airfoil.

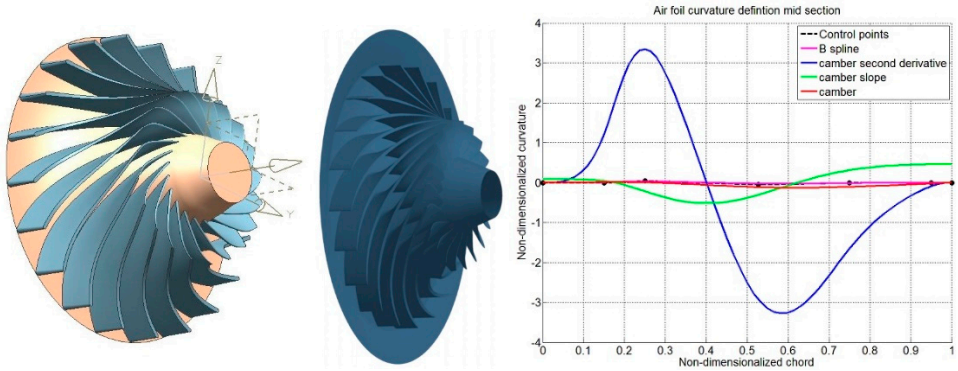



**Figure 25.** Centrifugal compressor design with curvature B-spline at midspan of S-shaped airfoil.

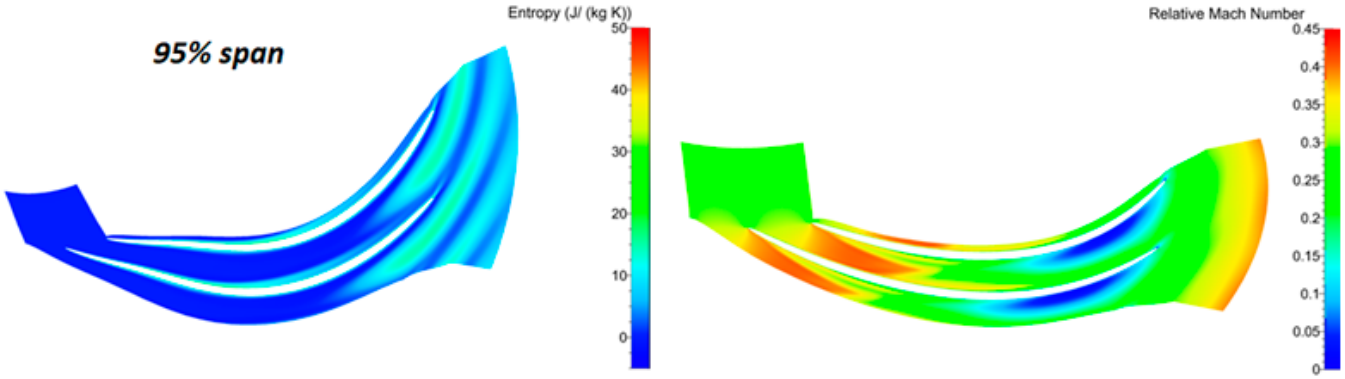



**Figure 26.** Contours of entropy and relative mach number at 95% span showing S-shaped profile.

The single-stage centrifugal compressor has a high pressure ratio compared with an axial compressor, and a multistage centrifugal compressor is not as efficient because the flow has to turn from radial at the outlet to axial at the inlet for the next stage. A novel multi-rotor centrifugal compressor on a single hub invented by Abdallah [84] is explored in collaboration with Mishra [83], Shaaban Abdallah and Sai Muppana [85] which utilizes the efficient axial and radial compressor flowpath as shown in Figure 27 by splitting the single rotor into an axial rotor, stator, and a purely radial rotor on a single disk.

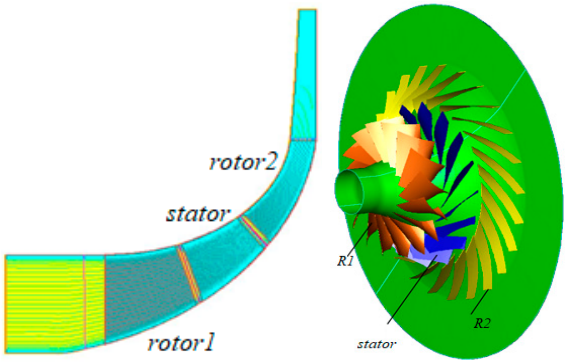



**Figure 27.** Novel 1.5 stage centrifugal compressor meridional view and 3D blade shape.

Two rotating rows of blades are mounted on the same impeller disk, separated by a stator blade row attached to the casing. This arrangement allows for more turning of the flow due to the presence of the stator which allows the second rotor to be independent of the first rotor orientation as described by Muppana et al. [85,86] in collaboration with the authors. Muppana [85] also developed a low-fidelity meanline design tool for centrifugal compressors with the authors to obtain spanwise metal angle distribution at LE and TE.

Single stage is designed for best angle distribution for pressure ratio of 4 and is used to create the novel 1.5 stage compressor as shown in Figure 27.

Figure 28 shows the azimuthal properties for single stage and the novel compressor. Relative mach number is lower and shows that there is a high enthalpy rise in the novel compressor and thus the size of the diffuser required further downstream will be much smaller. Several combinations of rotor–stator stages can be configured to maximize efficiency, pressure ratio, and surge margin.

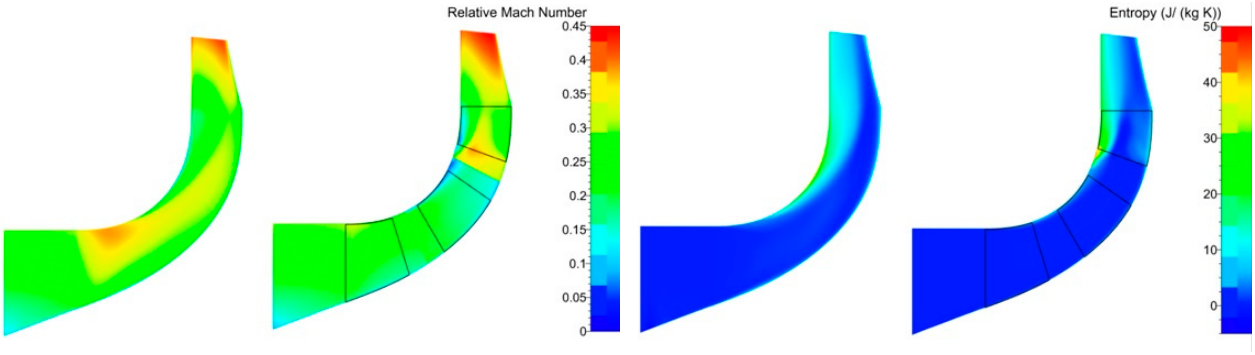



**Figure 28.** Mrel and entropy azimuthal contours of single and novel 1.5 stage compressor.

### 3.2.2. Turbopump, Curved LE Impeller, Radial De-Swirler, and Diffuser Vane

A representative liquid oxygen turbopump with three blades is generated with minimal parameters to demonstrate versatility of the tool as shown in Figure 29. Streamlines and parameters at hub and shroud stations are enough to create a baseline shape with interpolated values for radial stations between them. Curved LE radial impellers, de-swirler vanes are also easy to create using minimal parameters as shown in Figure 30.

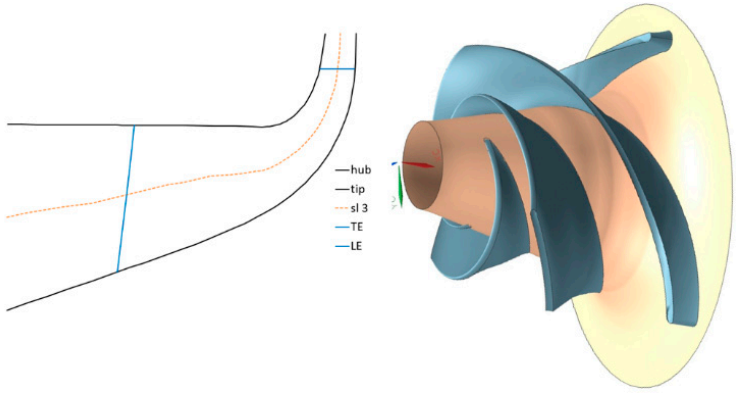



**Figure 29.** Representative turbopump generated with minimal parameters.

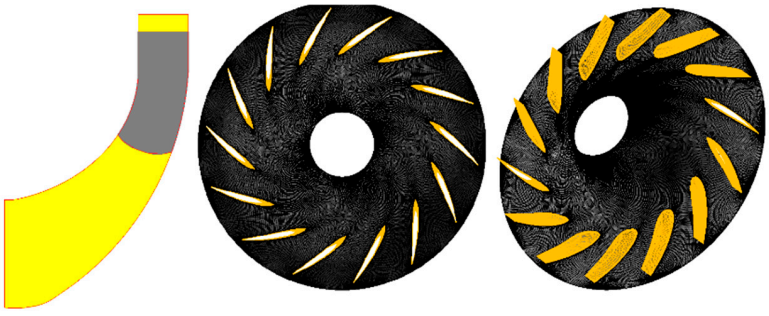


**Figure 30.** *Cont.*

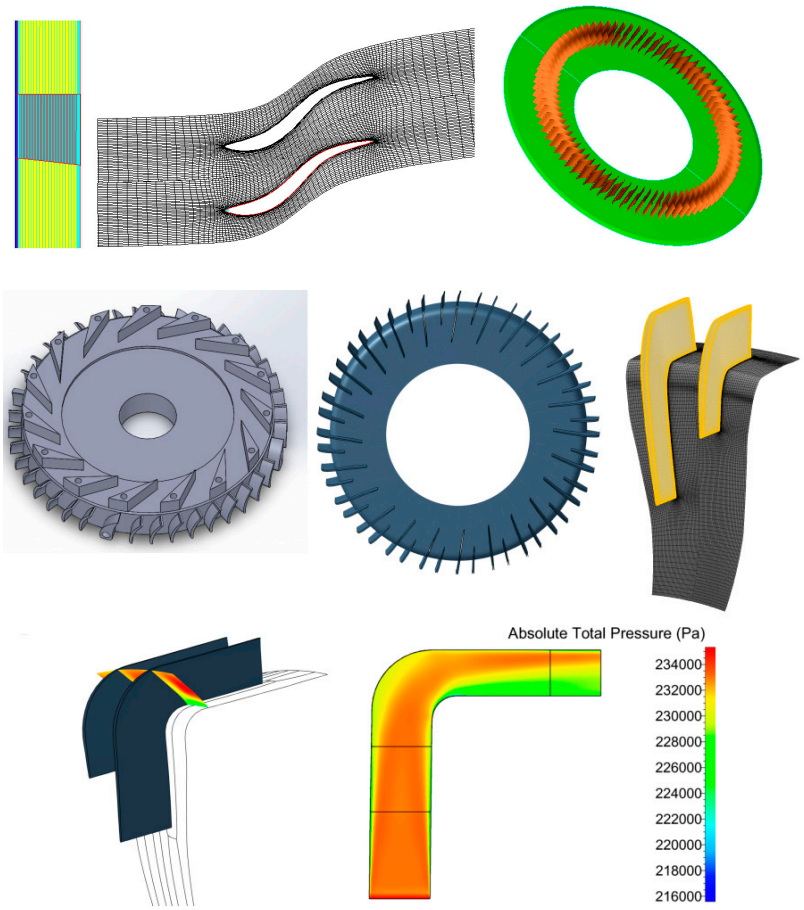



**Figure 30.** Curved leading edge design with few parameters, radial de-swirler vane, and a redesign of radial diffuser for JetCAT P90 engine.

Aero components of a JetCAT P90 gas turbine engine were redesigned for performance improvement and weight reduction by University of Cincinnati in collaboration with the Air Force Research Laboratory, Dayton, Ohio as part of Advanced Propulsion Outreach Program. A wedge-based radial diffuser was redesigned with a continuous vane with a splitter. LE and TE metal angles, end-wall contouring, position of the splitter, and lean were modified to achieve the blade shape shown in Figure 30.

### *3.3. Unducted Turbomachines*

Unducted rotors—including propellers, helicopters, wind, and hydrokinetic turbines—can also be generated using this tool. 2D airfoils can be imported or generated using the curvature option, spanwise stagger angle and chord are defined to obtain the 3D blade shape. A low-fidelity tool developed based on blade element momentum theory with physics enhancements provides the spanwise stagger and chord definitions are splined for smoothness.

#### 3.3.1. Propeller

A solo propeller blade is created and analyzed to demonstrate the dissipation of kinetic energy downstream. Spanwise chord and stagger angles are defined by the BEMT tool [48]. A NACA 4415 airfoil is used and Figure 31 shows the y+ is within an acceptable range. It also shows spanwise kinetic energy line plots at stations downstream of the rotor in a domain size of 15R upstream, 15R free stream, and 40R downstream. Figure 32 compares vorticity vectors on skin friction contours for the top 15% of the suction surface with several domains, turbulence transition models, and mesh density and their effect on prediction.

It has two blades, a 0.3 m hub, and a 1.75 m tip diameter; operating at 2400 RPM and 49.17 m/s at design inlet.

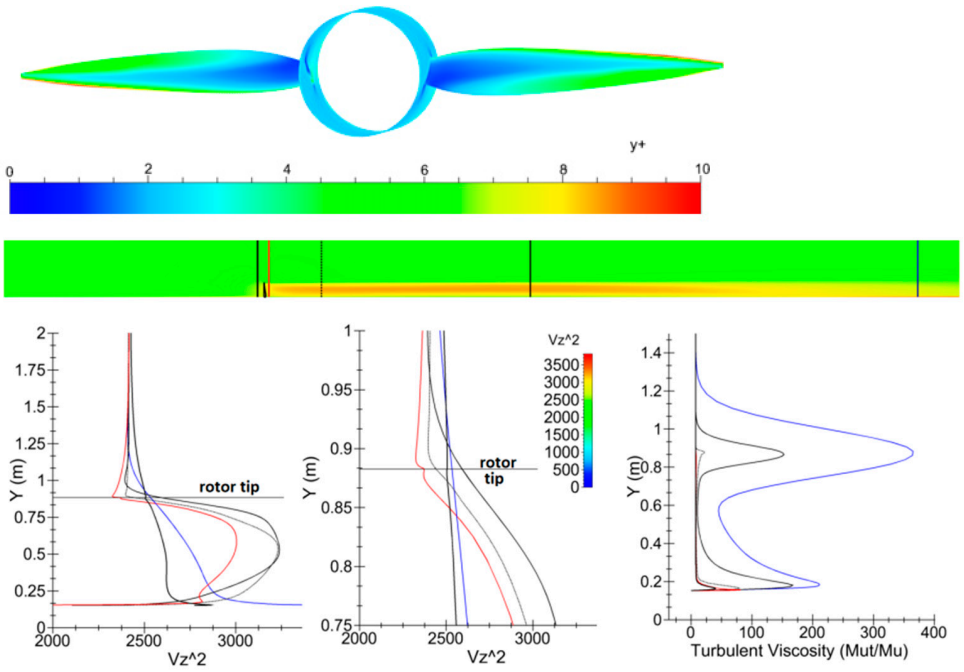



**Figure 31.** y+ and KE radial plots at several locations downstream of rotor in 3X-40R domain.

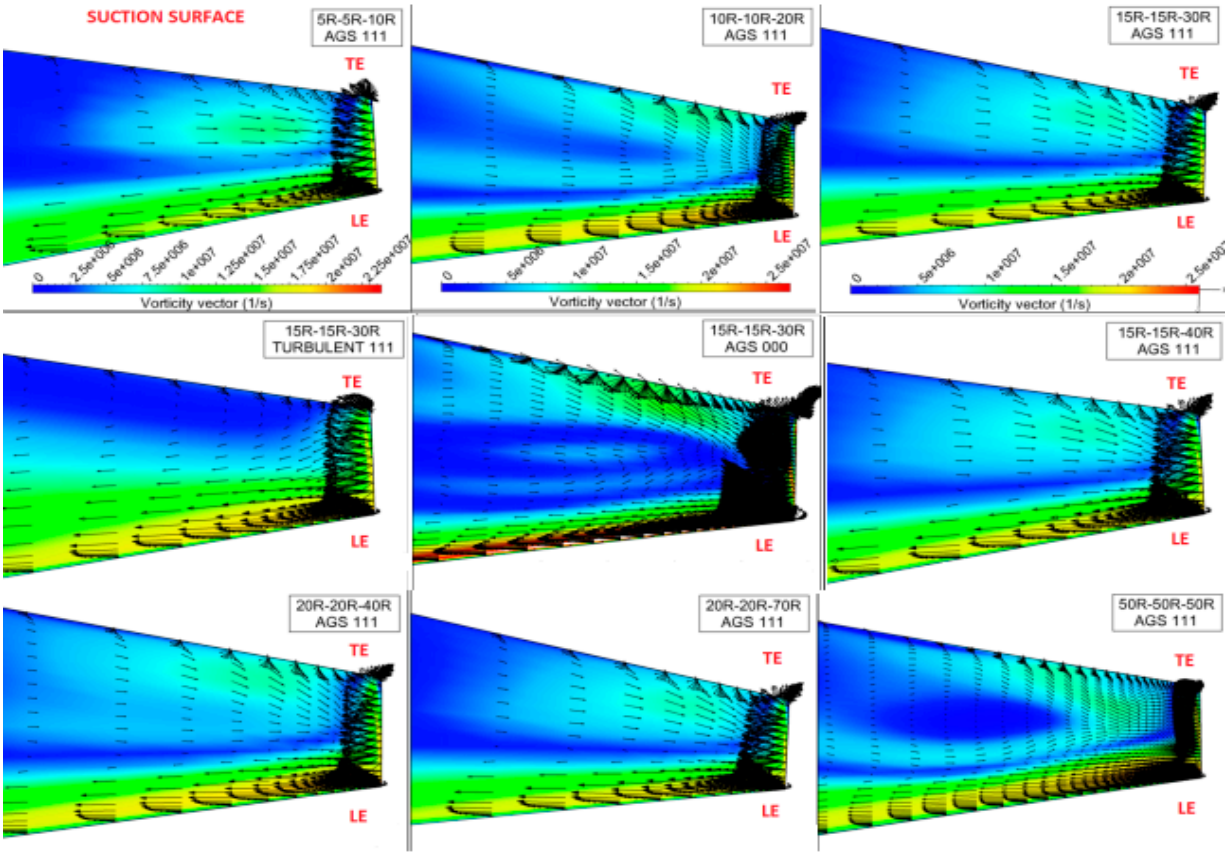



**Figure 32.** Vorticity vectors on skin friction contour for top 15% in various domains and transition models analyzed.

Swept propeller designed for a JetCat P80-SE outreach program with Air Force Research Laboratory (AFRL), Dayton, Ohio is also described. The goal was to design an exhaust driven fan of 12-inch diameter for the small P80-SE engine of 4.4-inch diameter, 2.9 pounds and producing a thrust of 22 pounds at 125,000 RPM. An ungeared 5 bladed swept propeller connected to the exhaust turbine and a transition duct attached aft of the nozzle was decided as the solution. Lean perturbation was used for sweep definition due to the high stagger angles of the propeller as shown in Figure 33. Figure 34 shows a supercritical NASA airfoil shape used near tip while keeping other spanwise airfoils as ClarkY to eliminate shocks near the tip section. This reduced the mach number and entropy. The device was built for testing at AFRL.

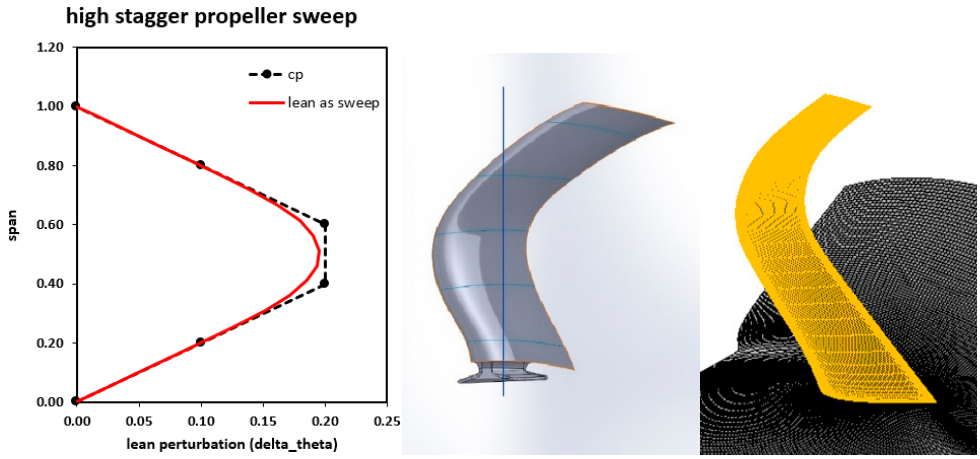



**Figure 33.** Highly staggered swept propeller with supercritical NASA airfoil at the tip.

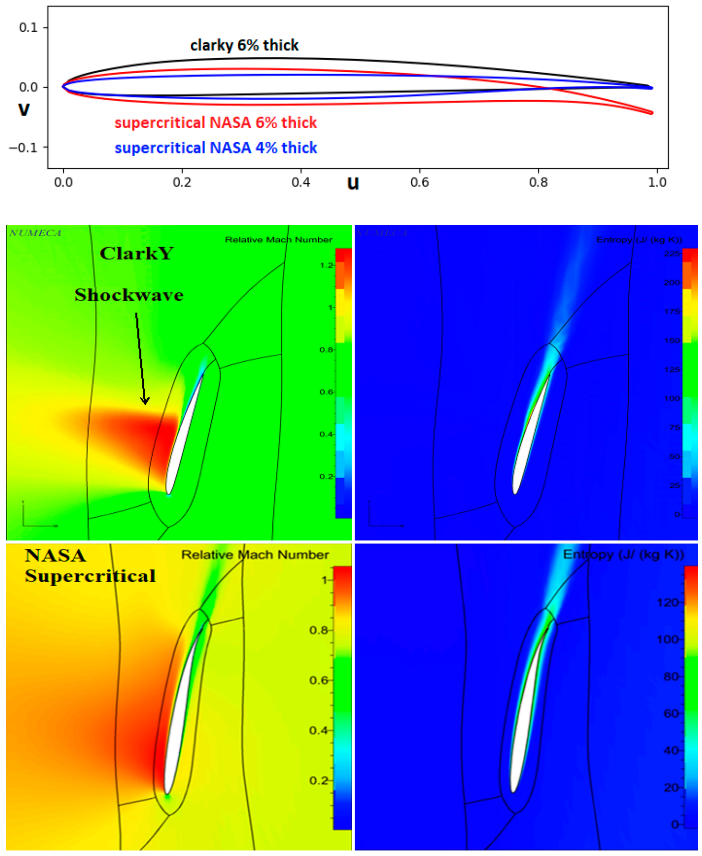



**Figure 34.** *Cont.*

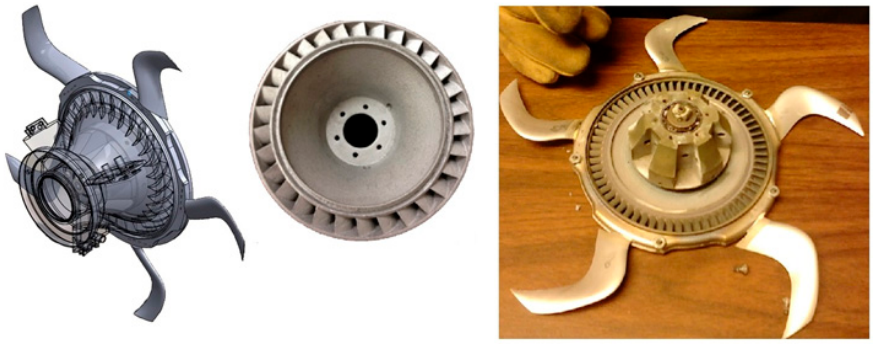


**Figure 34.** Swept propeller with supercritical NASA airfoil near the tip to avoid shockwave.

#### 3.3.2. Wind Turbine

Wind turbines can be analyzed and designed as high-aspect-ratio and highly staggered turbomachinery. Spanwise definition of stagger, normalized actual chord, and airfoils imported in ($u$, $v$) systems creates a smooth 3D blade. Hub to 25% of the span, blade-to-blade effects, or the cascade effect were considered, from 25% to 75%, the blade was designed as a 2D rotating wing and from 75% to the tip, 3D and tip effects were considered. A winglet tip version of the wind turbine is also generated through tangential lean. The lean is defined locally using a cubic B-spline with very few control points as shown in Figure 35. Since wind turbines are highly staggered, a translation of the blade section in the m′ direction corresponds to a lean and in θ direction corresponds to a sweep. The required stagger angle, height of the winglet, sweep and chord can be controlled very easily. Only a pure lean in the downstream direction is explored here. The NREL Phase VI representative wind turbine with two blades and a diameter of 10.05 m is chosen for design and analysis which uses S809 airfoil created by NREL to have a maximum lift coefficient. The winglet tip reduces the induced angle of attack above 80% span which reduces the downwash because of the weakening of the tip vortices due to the production of additional thrust in the circulation field. Increase in the effective angle of attack above 80% span results in an improved lift coefficient. Torque and hence power output is also increased because of the downstream winglet tip. Figure 35 shows the streamtubes over standard and winglet tip wind turbine, demonstrating the tip vortices start developing in the standard blade but are delayed in winglet tip blade. The winglet tip alone increased the power by 2.63% from 4.7781 kW to 4.9037 kW purely due to aerodynamic improvement by reducing the complex 3D tip effects.

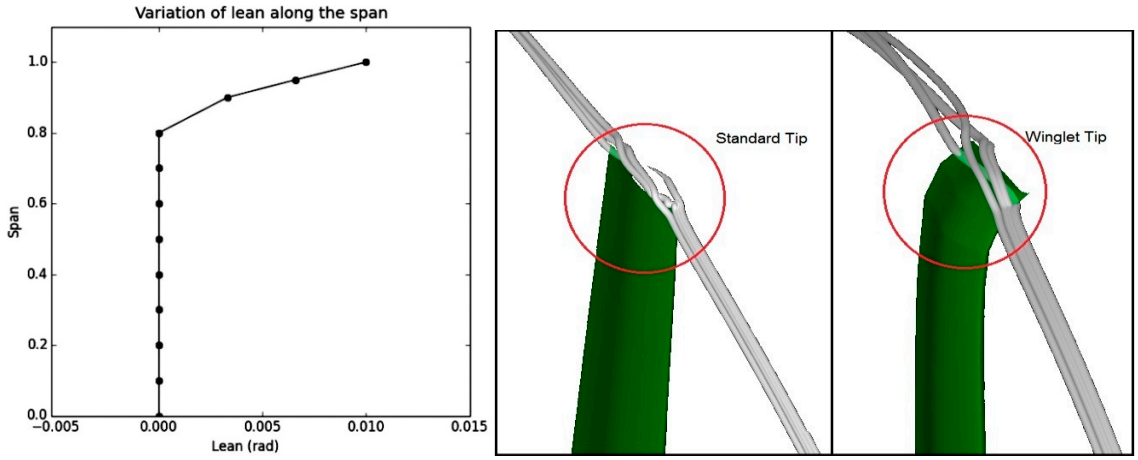



**Figure 35.** Spanwise local lean defined near the tip with smoother streamlines for winglet blade.

Several blade shapes with forward and backward sweep and forward and backward translation were analyzed using a coning angle of 20 degrees with S809 airfoil. After a proof

of concept by Siddappaji, Guruprasad et al. [87] conducted analysis of these configurations in collaboration with authors as part of a class project. Figure 36 shows all the configurations created and analyzed in 3D CFD. Figure 37 shows axisymmetric view of axial velocity distributions, showing a potential for hurricane resistant wind turbine blades and the backswept blade is evidently a better choice.

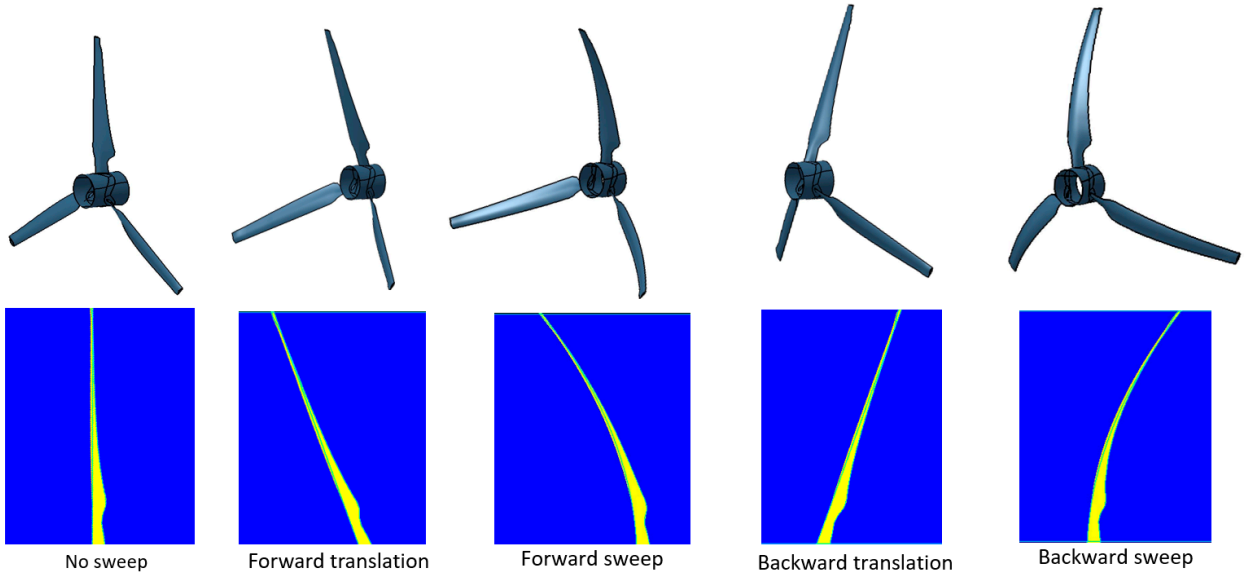



**Figure 36.** Different definitions of sweep applied to a wind turbine rotor.

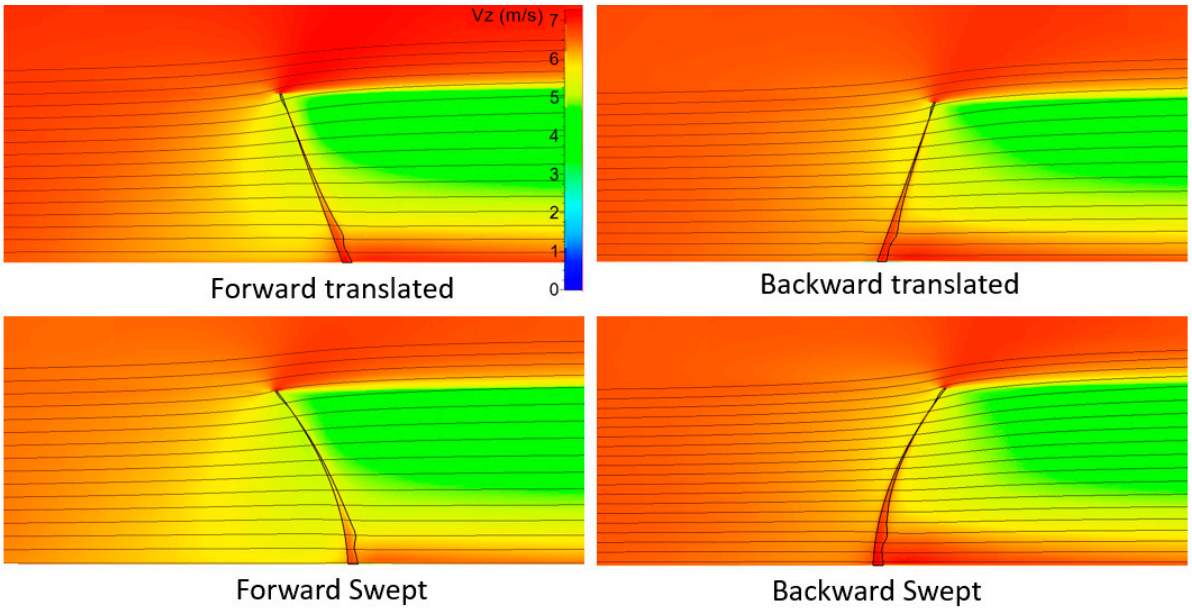



**Figure 37.** Axial velocity in axisymmetric view for different sweep and translation definitions.

### 3.3.3. Hydrokinetic Turbine

Hydrokinetic turbines convert kinetic energy of moving water to electric power without a pressure head. Spanwise definition of chord and stagger obtained from a low-fidelity BEMT tool [48] are used as inputs to create 3D blade shape as shown in Figure 38. The Eppler 857 hydrofoil is a better option than the traditional S809 airfoil for such turbines as shown by the lift and drag properties from Xfoil [88]. It also shows laminar-to-turbulent transition chord locations for top and bottom surfaces at various angles of attack, $\alpha$. The

turbine has a 0.45 m tip radius and 0.1 m hub radius, spinning at 97 RPM and inlet velocity of 1.5 m/s.

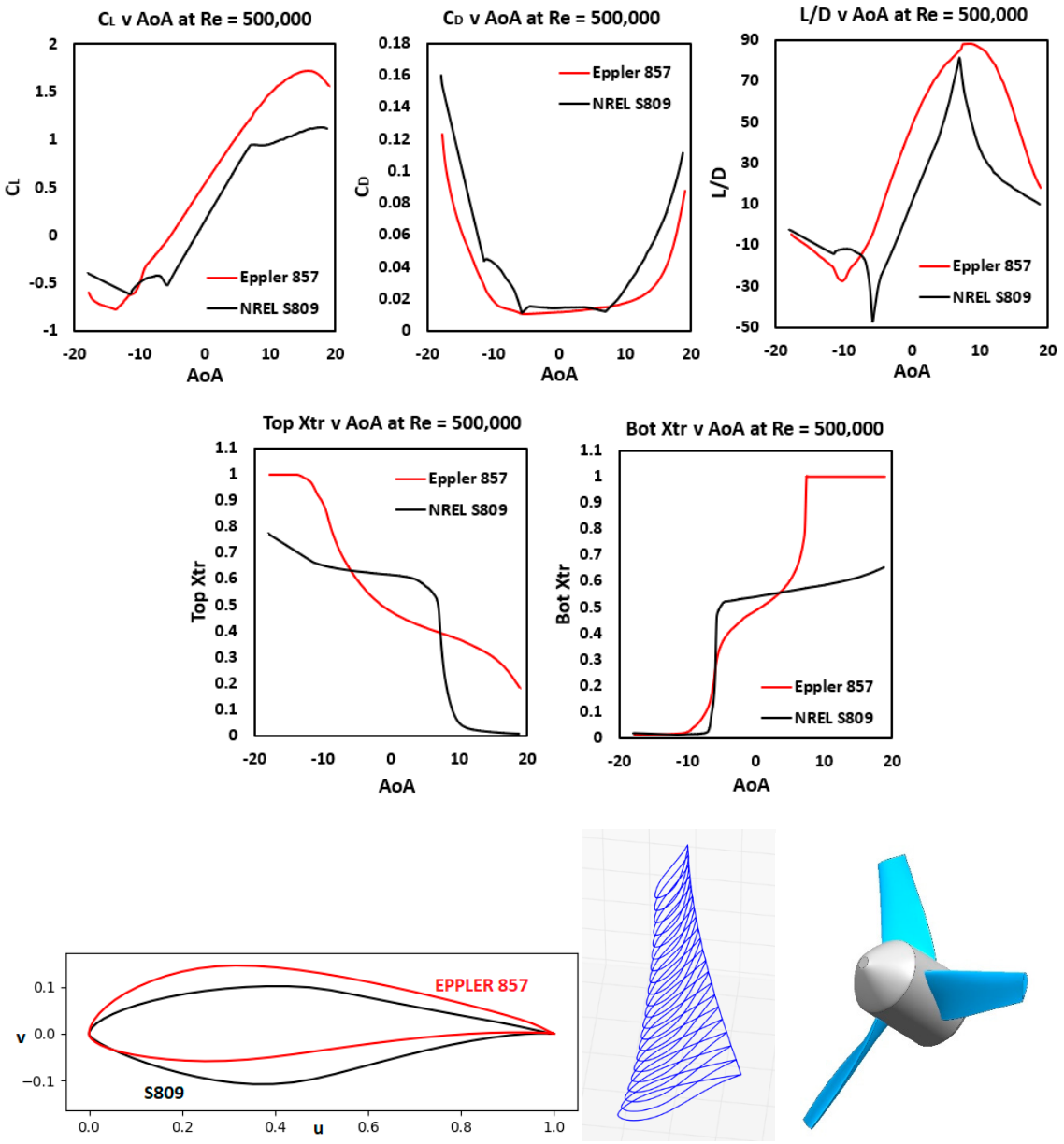



**Figure 38.** Eppler 857 hydrofoil compared with S809 and 3D blade for hydrokinetic turbine.

## 4. Conclusions

A versatile parametric design tool for 3D blade design of turbomachinery has been developed and improved over a period of several years. Capability to construct 3D blades for all types of turbomachinery demonstrates generality through various applications. The tool outputs blade shape in several data files containing specified coordinates of 3D blade sections which can be imported into a CAD package to obtain a smooth lofted blade. Parametric definition makes the geometry modification process more cost-effective, quicker, and easier. A 2D curvature-defined meanline blade airfoil geometry generator creates a fifth order meanline by twice integrating a cubic B-spline that describes a second derivative and allows curvature and slope-of-curvature to be continuous parametrically, C2, which is better than G2 continuity. Applying an adequate shape control of the blade surface reduces spikes in mach number and pressure distributions. Controlling thickness distribution parametrically using fourth order B-spline also adds smoothness to the shape. Global

and local sweep and lean manipulation defined by B-splines create winglet tips and split tips which minimize loss due to tip leakage and secondary flows. Spanwise manipulation of curvature, thickness, and other parametric properties smoothly using cubic B-splines have been implemented and are useful in designing efficient 3D blades. Insertion ability of general 2D airfoils at any span, and consistent 3D stacking in a rigorous manner, sets this blade generator apart. Simplicity of the tool makes it easy to implement as part of an automated chain and renders itself as a powerful design package to solve complicated aerodynamic-performance-driven problems when tied to a flow solver. Python plotters of 2D sectional passage, 3D blade view, curvature, connection to CAD, and CFD are all part of the package. Robustness is evident from the variety of applications demonstrated. The latest executable and data plotters are freely available at http://gtsl.ase.uc.edu/t-blade3/ (accessed on 20 July 2022) and source code at https://github.com/GTSL-UC/T-Blade3 (accessed on 20 July 2022).

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**Conflicts of Interest:** The authors declare no conflict of interest.

## Nomenclature

| | |
|---|---|
| a | Percent offset |
| b | Intermediate scaling factor |
| B | Basis function |
| $C_L$, $C_D$ | Coefficient of lift and drag |
| C | Curvature, parametric curve continuity |
| G | Geometric continuity |
| L | Reference length |
| m | Meridional |
| P | Control point |
| R | Radius of curvature, rotor |
| r | Radial coordinate |
| S | B-spline segment |
| s | Entropy |
| scf | Dimensional blade scaling factor |

| | |
|---|---|
| t | Parameter t (0, 1) for spline construction |
| thk | Thickness definition as a function of u |
| u | Normalized chordwise coordinate (0,1) |
| v | Meanline coordinates as a function of u |
| V | Vane |
| x, y, z | Cartesian coordinates |
| y+ | Non-dimensional wall distance |
| $\alpha$ | Angle of attack |
| $\Omega$ | Rotational speed of the rotor |
| $\beta$, $\beta^*$ | Flow angle, metal angle |
| $\varphi$ | Slope of streamline |
| $\zeta$ | Stagger angle |
| $\theta$ | Tangential coordinate |
| AGS | Abu-Ghannam/Shaw |
| BEMT | Blade element momentum theory |
| CAD | Computer-aided design |
| CFD | Computational fluid dynamics |
| $E^3$ | Energy efficient engine |
| HPC | High-pressure compressor |
| KE | Kinetic energy |
| LE | Leading edge |
| LPC | Low pressure compressor |
| NACA | National Advisory Committee for Aeronautics |
| NASA | National Aeronautics and Space Administration |
| NREL | National Renewable Energy Laboratory |
| OGV | Outlet guide vane |
| RANS | Reynolds averaged Navier–Stokes |
| TE | Trailing edge |
| 2D, 3D | Two- and three-dimensional |

## Appendix A

A flowchart for generating various 2D airfoils in the tool is described in Figure A1. Curvature of meanline driven shapes is separated from external airfoils or other type of airfoil definitions embedded in the tool. A flowchart for the optimization process and geometry generator connected to a CFD solver and finite element analysis tool at various fidelities is shown in Figure A2.

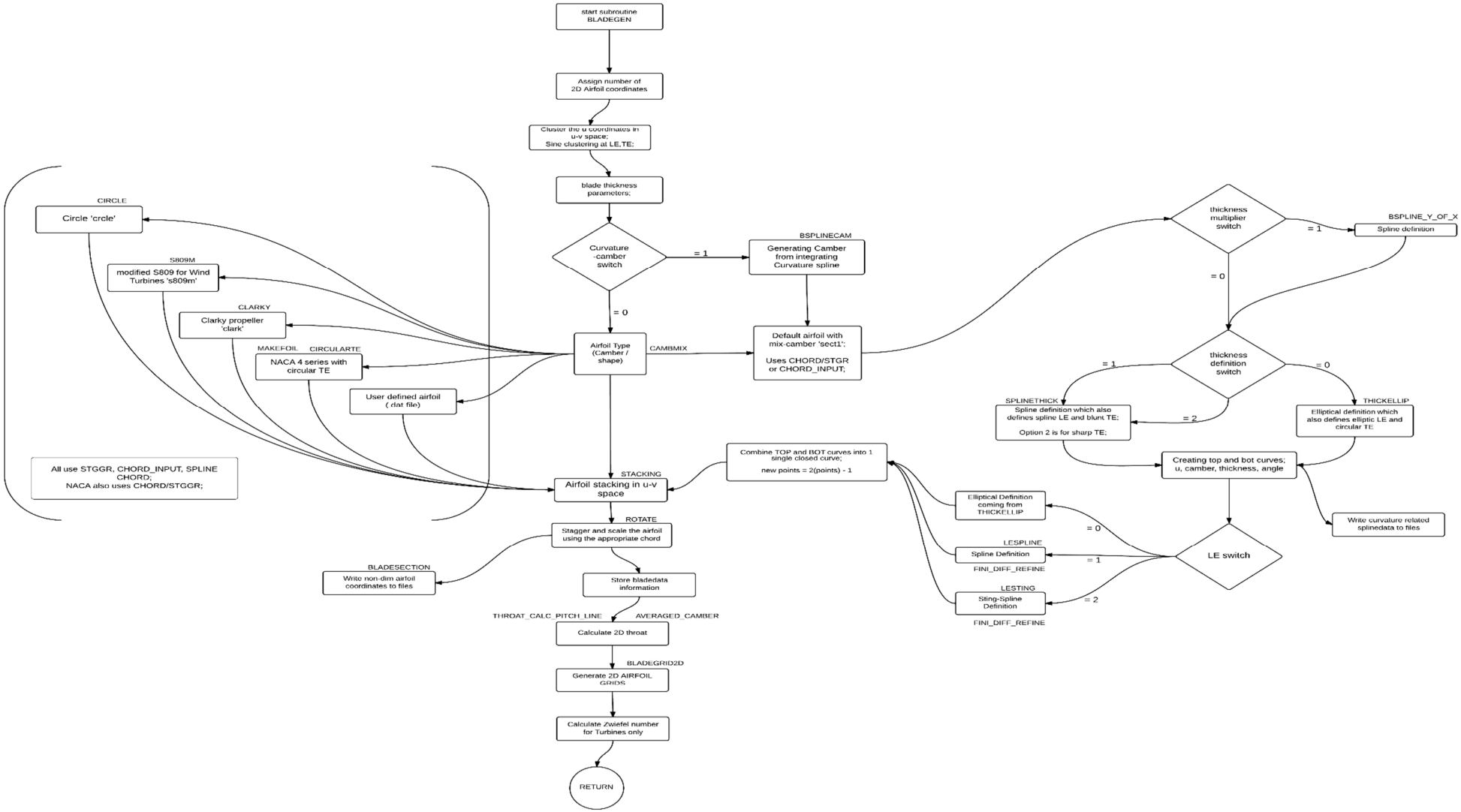


**Figure A1.** Algorithm for generating 2D airfoils with various options in the tool.

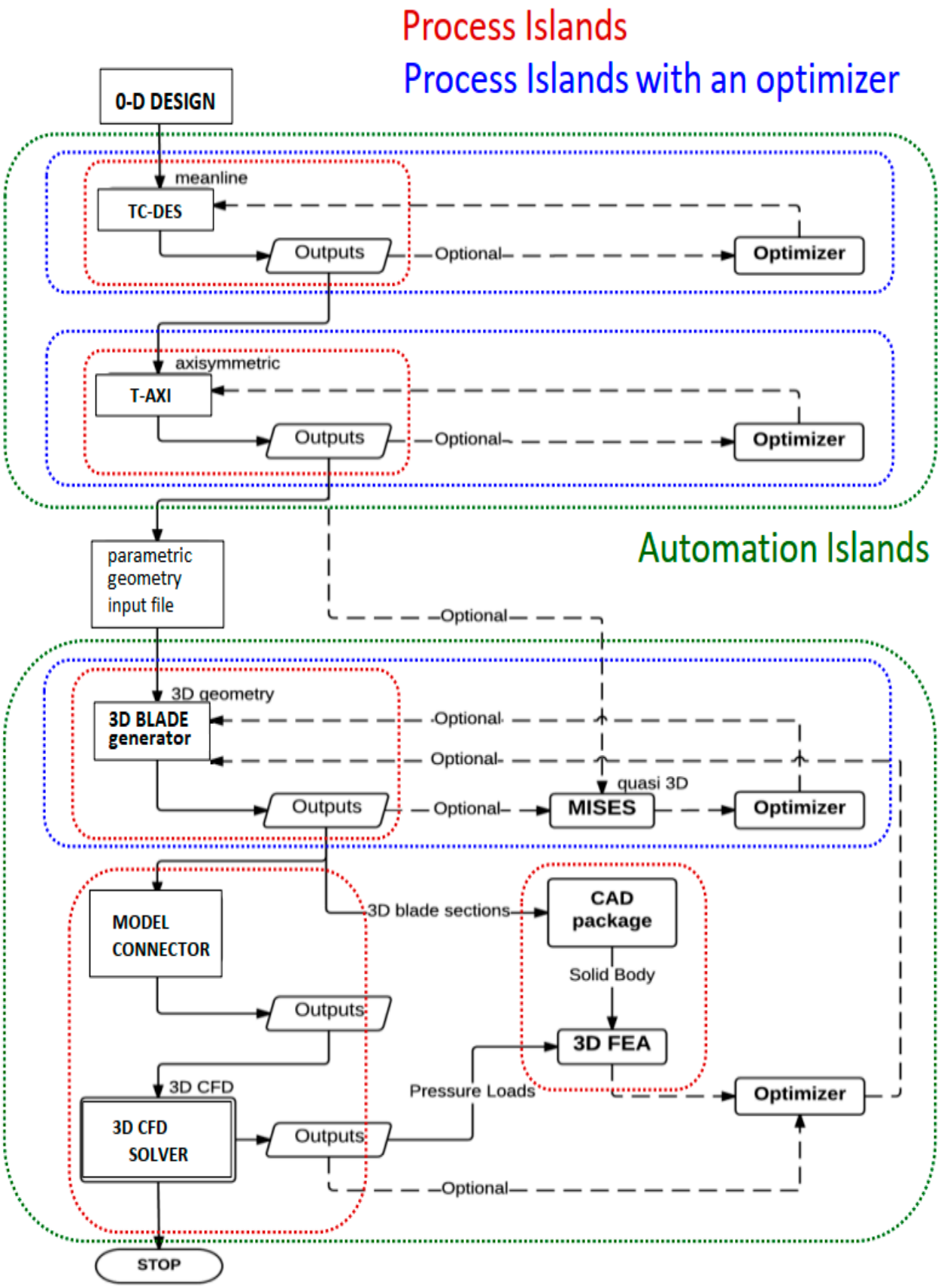


**Figure A2.** Automated framework integrated with an optimizer for multiphysics.

## References


1. Miller, P.L.; Oliver, J.H.; Miller, D.P.; Tweet, D.L. Bladecad: An interactive geometric design tool for turbomachinery blades. In Proceedings of the 41st Gas Turbine and Aeroengine Congress, ASME, Birmingham, UK, 10–13 June 1996. NASA Technical Memorandum 107262.
2. Kulfan, B.M. A Universal Parametric Geometry Representation Method—"CST". AIAA 2007-62. 07. In Proceedings of the 45th AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, USA, 8–11 January 2007.
3. Kulfan, B.M.; Bussoletti, J.E. Fundamental Parametric Geometry Representations for Aircraft Component Shapes. AIAA 2006-6948. In Proceedings of the 11th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference, Portsmouth, VA, USA, 6–8 September 2006.
4. Kulfan, B.M. Universal Parametric Geometry Representation Method. *J. Aircr.* **2008**, *45*, 142–158. [CrossRef]
5. Sripawadkul, V.; Padulo, M.; Guenov, M. A comparison of airfoil shape parameterization techniques for early design optimization. In Proceedings of the 13th AIAA/ISSMO Multidisciplinary Analysis Optimization Conference, Fort Worth, TX, USA, 13–15 September 2010.
6. Koini, G.N.; Sarakinos, S.S.; Nikolos, I.K. A software tool for parametric design of turbomachinery blades. *Adv. Eng. Softw.* **2009**, *40*, 41–51. [CrossRef]
7. Crouse, J.E.; Gorrell, W.T. *Computer Program for Aerodynamic and Blading Design of Multistage Axial-Flow Compressors*; Tech. Rep. TP 1946; NASA Lewis Research Center: Cleveland, OH, USA, 1981; pp. 1–110.
8. Engeli, M.; Zollinger, H.J.; Allemann, J.C. A computer program for the design of turbomachinery blades. In *Turbo Expo: Power for Land, Sea, and Air*; ASME: New York, NY, USA, 1974; pp. 1–10. [CrossRef]
9. Casey, M.V. A computational geometry for the blades and internal flow channels of centrifugal compressors. *Trans. ASME J. Eng. Power.* **1983**, *105*, 288–295. [CrossRef]
10. Pierret, S.; Demeulenaere, A.; Gouverneur, B.; Braembussche, V.R. Designing turbomachinery blades with the function approximation concept and the Navier-Stokes equations. In Proceedings of the 8th Symposium on Multidisciplinary Analysis and Optimization, Long Beach, CA, USA, 6–8 September 2000; pp. 1–11. [CrossRef]
11. Oyama, A.; Liou, M.S.; Obayashi, S. Transonic axial-flow blade optimization: Evolutionary algorithms/three-dimensional Navier-Stokes solver. *J. Propuls. Power* **2004**, *20*, 612–619. [CrossRef]
12. Verstraete, T. CADO: A computer aided design and optimization tool for turbomachinery applications. In Proceedings of the 2nd International Conference on Engineering Optimization, Lisbon, Portugal, 6–9 September 2010; pp. 1–10.
13. Müller, L.; Verstraete, T. CAD Integrated multipoint adjoint-based optimization of a turbocharger radial turbine. *Int. J. Turbomach Propuls. Power* **2017**, *2*, 14. [CrossRef]
14. Leylek, Z.; Neely, A.J. Development of a blade parametric modeling methodology for design and analysis of computer experiments. In Proceedings of the ASME Turbo Expo GT2015-43109, Montreal, Quebec, Canada, 15–19 June 2015.
15. Korakianitis, T.P. Design of airfoils and cascades of airfoils. *AIAA J.* **1989**, *27*, 455–461. [CrossRef]
16. Korakianitis, T.P.; Pantazopoulos, G.I. Improved turbine-blade design techniques using 4th-order parametric-spline segments. *Comput. Aided Des.* **1993**, *25*, 289–299. [CrossRef]
17. Korakianitis, T.P.; Hamakhan, I.A.; Resaienia, M.A.; Wheeler, A.P.S.; Avital, E.J.; Williams, J.J.R. Design of high-efficiency turbomachinery blades for energy conversion devices with the three-dimensional prescribed surface curvature distribution blade design (circle) method. *Appl. Energy* **2012**, *89*, 215–227. [CrossRef]
18. Burman, J. Geometry Parameterisation and Response Surface-Based Shape Optimization of Aero Engine Compressors. Ph.D. Thesis, Lulea University of Technology, Norrbotten, Sweden, 2003.
19. Hamakhan, I.A.; Korakianitis, T.P. Aerodynamic performance effects of leading-edge geometry in gas-turbine blades. *Appl. Energy* **2010**, *87*, 1591–1601. [CrossRef]
20. Song, Y.; Gu, C. Effects of Curvature Continuity of Compressor Blade Profiles on Their Performances. In Proceedings of the ASME Turbo Expo 2014, Düsseldorf, Germany, 16–20 June 2014.
21. Bi, Q.; Chen, H.; Tong, D.; Lu, Y.; Zou, X. Design Method and Performance Effects of Curvature-Smooth Centrifugal Compressor Blades. In Proceedings of the ASME Turbo Expo 2015, Montreal, Canada, 15–19 June 2015.
22. Meckstroth, C.; Brown, J. Point Cloud to Parameter: An Inverse Geometric Approach to Probabilistic Design. In Proceedings of the ASME Turbo Expo 2019, Phoenix, AZ, USA, 17–21 June 2019.
23. McKeand, A.M.; Gorguluarslan, R.M.; Choi, S. A Stochastic Approach for Performance Prediction of Aircraft Engine Components Under Manufacturing Uncertainty. In Proceedings of the ASME 2018 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, Quebec, Canada, 26–29 August 2018.
24. Benini, E. Three-dimensional multi-objective design optimization of a transonic compressor rotor. *J. Propuls. Power* **2004**, *20*, 462–559. [CrossRef]
25. Lian, Y.; Liou, M.S. Multi-objective optimization of transonic compressor blade using evolutionary algorithm. *J. Propuls. Power* **2005**, *21*, 979–987. [CrossRef]
26. Ellbrant, L.; Eriksson, L.E.; Martenssion, H. Balancing efficiency and stability in the design of transonic compressor stages. In Proceedings of the ASME Turbo Expo GT2013-94838, San Antonia, Texas, USA, 3–7 June 2013.
27. Zhang, X.; Qiang, X.; Teng, J.; Yu, W. A new curvature-controlled stacking-line method for optimization design of compressor cascade considering surface smoothness. *Proc. Inst. Mech. Eng. Part G J. Aerosp. Eng.* **2020**, *234*, 1061–1074. [CrossRef]

28. Sharma, M.; Turner, M.G. Continuous Modified NACA Four-Digit Thickness Distribution for Turbomachinery Geometries. *AIAA J.* **2021**, *59*, 1501–1505. [CrossRef]
29. Dannenhoffer, J.F.; Haimes, R. Generation of Multi-fidelity, Multi-discipline Air Vehicle Models with the Engineering Sketch Pad. In Proceedings of the 54th AIAA Aerospace Sciences Meeting, AIAA, 6.2016-1925, San Diego, CA, USA, 4–8 January 2016.
30. Dannenhoffer, J.F.; Haimes, R. Using Design-Parameter Sensitivities in Adjoint-Based Design Environments. In Proceedings of the 55th AIAA Aerospace Sciences Meeting, AIAA 2017-0139, Grapevine, TX, USA, 9–13 January 2017.
31. Canfield, R.A.; Alnaqbi, S.; Durscher, R.J.; Bryson, D.E.; Kolonay, R.M. Shape Continuum Sensitivity Analysis using ASTROS and CAPS. In Proceedings of the AIAA Scitech Forum, AIAA, 6.2019-2228, San Diego, CA, USA, 7–11 January 2019.
32. Meckstroth, C.M. Parameterized, Multi-fidelity Aircraft Geometry and Analysis for MDAO Studies using CAPS. In Proceedings of the AIAA Scitech 2019 Forum, AIAA, 6.2019-2230, San Diego, CA, USA, 7–11 January 2019.
33. Agromayor, R.; Anand, N.; Müller, J.-D.; Pini, M.; Nord, L. A Unified Geometry Parametrization Method for Turbomachinery Blades. *Comput. Aided Des.* **2021**, *133*, 102987. [CrossRef]
34. Luo, S.; Cui, J.; Sella, V.; Liu, J.; Koric, S.; Kindratenko, V. Turbomachinery Blade Surrogate Modeling using Deep Learning. In *ISC High Performance*; Springer: Cham, Switzerland, 2021; Lecture Notes in Computer Science 12761.
35. Shi, H. A Parametric Blade Design Method for High-Speed Axial Compressor. *Aerospace* **2021**, *8*, 271. [CrossRef]
36. Shi, H. Parametric Research and Aerodynamic Characteristic of a Two-Stage Transonic Compressor for a Turbine Based Combined Cycle Engine. *Aerospace* **2022**, *9*, 346. [CrossRef]
37. Yang, J.; Hu, B.; Tao, Y.; Li, J. A flexible method for geometric design of axial compressor blades. *Proc. Inst. Mech. Eng. Part G J. Aerosp. Eng.* **2022**, *236*, 2420–2432. [CrossRef]
38. Sairam, K. The Influence of Radial Area Variation on Wind Turbines to the Axial Induction Factor. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2013.
39. Brossman, J.R. An Investigation of Rotor Tip Leakage Flows in the Rear-Block of a Multistage Compressor. Ph.D. Thesis, Purdue University, Pennsylvania, PA, USA, 2012.
40. Srinivas, A. Novel Compressor Blade Design Study. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2015.
41. Syed, M.H.M. Optimization Capabilities for Axial Compressor Blades and Seal Teeth Cavity. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2016.
42. Masters, I.; Williams, A.; Croft, T.; Togneri, M.; Edmunds, M.; Zangiabadi, E.; Fairley, I.; Karunarathna, H. A Comparison of Numerical Modelling Techniques for Tidal Stream Turbine Analysis. *Energies* **2015**, *8*, 7833–7853. [CrossRef]
43. Wu, Y.T.; Porte-Agel, F. Atmospheric Turbulence Effects on Wind-Turbine Wakes: An LES Study. *Energies* **2012**, *5*, 5340–5362. [CrossRef]
44. Bhide, K.; Siddappaji, K.; Abdallah, S.; Roberts, K. Improved Supersonic Turbulent Flow Characteristics Using Non-Linear Eddy Viscosity Relation in RANS and HPC-Enabled LES. *Aerospace* **2021**, *8*, 352. [CrossRef]
45. Bhide, K.; Abdallah, S. Turbulence statistics of supersonic rectangular jets using Reynolds Stress Model in RANS and WALE LES. In Proceedings of the AIAA AVIATION 2022 Forum, AIAA 2022-3344, Chicago, IL, USA, 27 June–1 July 2022.
46. Bhide, K.; Abdallah, S. Anisotropic Turbulent Kinetic Energy budgets in compressible rectangular jets. *Aerospace* **2022**, *9*, 484. [CrossRef]
47. Turner, M.G.; Bruna, D.; Merchant, A. Applications of a turbomachinery design tool for compressors and turbines. In Proceedings of the 43rd AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit, Cincinnati, OH, USA, 8–11 July 2007; pp. 2007–5152.
48. Siddappaji, K. On the Entropy Rise in General Unducted Rotors Using Momentum, Vorticity and Energy Transport. Ph.D. Thesis, University of Cincinnati, Cincinnati, OH, USA, 2018.
49. Denton, J.D. The effects of lean and sweep on transonic fan performance: A computational study. In *Turbo Expo: Power of Land Sea and Air*; ASME: New York, NY, USA, 2002.
50. Smith, J.; Leroy, H.; Yeh, H. Sweep and dihedral effects in axial-flow turbomachinery. *J. ASME* **1963**, *85*, 401–414. [CrossRef]
51. Siddappaji, K. Parametric 3d Blade Geometry Modeling Tool for Turbomachinery Systems. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2012.
52. Wennerstrom, A.J. *Design of Highly Loaded Axial-fow Fans and Compressors*; Concepts ETI, Inc.: Windsor, VT, USA, 2000.
53. Nemnem, A.F.; Turner, M.G.; Siddappaji, K.; Galbraith, M.C. A smooth curvature-defined meanline section option for a general turbomachinery geometry generator. In Proceedings of the ASME Turbo Expo 2014, Dusseldorf, Germany, 16–20 June 2014.
54. Sharma, M.; Dannenhoffer, J.; Holder, J.; Turner, M.G. Integration of an Open Source Blade Geometry Generator Using a Physics Based Parameterization with the Engineering Sketch Pad. In Proceedings of the ASME Turbo Expo 2020, Virtual, 21–25 September 2020. [CrossRef]
55. Turner, M.G.; Park, K.; Siddappaji, K.; Dey, S.; Gutzwiller, D.P.; Merchant, A.; Bruna, D. Framework for Multidisciplinary Optimization of Turbomachinery. In Proceedings of the ASME Turbo Expo 2010, Glasgow, UK, 14–18 June 2010; pp. 623–631.
56. Goodhand, M.N.; Miller, R.J. The impact of real geometries on three-dimensional separations in compressors. In Proceedings of the ASME Paper Number GT2010-22246, Glasgow, UK, 14-18 June 2010.
57. Vince, J. *Mathematics for Computer Graphics*, 2nd ed.; Springer: London, UK, 2006.
58. Hoffmann, I.J.M. On the quartic curve of han. *J. Comput. Appl. Math.* **2009**, *223*, 124–132.
59. Fortin, D. B-spline toeplitz inverse under corner perturbations. *Int. J. Pure Appl. Math.* **2012**, *77*, 107–118.
60. Sherar, P.A. Variational Based Analysis and Modelling Using B-Splines. Ph.D. Thesis, Cranfield University, Cranfield, UK, 2004.

61. Papp, I.; Hoffmann, M. C2 and G2 continuous spline curves with shape parameters. *J. Geom. Graph.* **2007**, *11*, 1–7.
62. Spekreijse, S.; Boerstoel, J.W. *Multiblock Grid Generation. Part I: Elliptic Grid Generation Methods for Structured Grids*; NLR TP 96338; National Aerospace Laboratory NLR: Amsterdam, The Netherlands, 1996.
63. Bauinger, S.; Gannon, A.J.; Hobson, G.V.; Terrell, A.D. Technology Demonstration of a Splittered Transonic Rotor with a Downstream Variable Geometry Tandem Stator. In Proceedings of the ASME Turbo Expo 2017, Charlotte, NC, USA, 26–30 June 2017.
64. Galbraith, M.C. A Discontinuous Galerkin Chimera Overset Solver. Ph.D. Theses, University of Cincinnati,, Cincinnati, OH, USA, 2013.
65. Knapke, R.D. High-Order Unsteady Heat Transfer with the Harmonic Balance Method. Ph.D. Theses, University of Cincinnati, Cincinnati, OH, USA, 2015.
66. Neshat, M.A.; Akhlaghi, M.; Fathi, A.; Khaledi, H. Investigating the effect of blade sweep and lean in one stage of an industrial gas turbine's transonic compressor. *Propuls. Power Res.* **2015**, *4*, 221–229. [CrossRef]
67. Eldred, M.S.; Adams, B.M.; Haskell, K.; Bohnhoff, W.J.; Eddy, J.P.; Gay, D.M.; William, E.H.; Hough, P.D.; Kolda, T.G.; Swiler, L.P.; et al. *DAKOTA Reference Manual*, 4.2 ed.; Sandia National Laboratories: Albuquerque, NM, USA, 2008.
68. Fineturbo/Autogrid5, Numeca. Available online: http://www.numeca.com/en/products/finetmturbo (accessed on 24 December 2020).
69. CD-Adapco. Starccm+ User Guide. Available online: https://www.plm.automation.siemens.com/global/en/products/simcenter/STAR-CCM.html (accessed on 24 December 2020).
70. Nemnem, A. A General Multidisciplinary Turbomachinery Design Optimization System Applied to a Transonic Fan. Ph.D. Theses, University of Cincinnati, Cincinnati, OH, USA, 2014.
71. Clark, T. Pressure Mapping and Efficiency Analysis of an EPPLER 857 Hydrokinetic Turbine. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2017.
72. Holloway, P.R.; Knight, G.L.; Koch, C.C.; Shaffer, S.J. *Energy Efficient Engine High Pressure Compressor Detail Design Report*; Technical Report; NASA-CR-165558; General Electric Company: Boston, MA, USA, 1982.
73. Claus, R.W.; Beach, T.; Turner, M.G.; Siddappaji, K.; Hendricks, E.S. Geometry and simulation results for a gas turbine representative of the energy efficient engine (EEE). Technical report; NASA: Washington, DC, USA, 2015.
74. Matthys, R.; Mesward, D.; Turner, M.G.; Siddappaji, K. *$E^3$ 6 Stage Simulation*; Turbomachinery Flow Class Project Internal Report; University of Cincinnati: Cincinnati, OH, USA, 2012.
75. Cherry, D.G.; Lenahan, D.T. *Energy Efficient Engine. Low Pressure Turbine Test Hardware Detailed Design Report. NASA CR 167956*; NASA: Washington, DC, USA, 1982.
76. Srinivas, A.; Siddappaji, K.; Turner, M.G. *Novel Compressor Blade Design Study*; International Society for Air Breathing Engines: Phoenix, AZ, USA, 2015; ISABE-2015-20263.
77. Kumar, S.; Turner, M.G.; Siddappaji, K.; Celestina, M. Aerodynamic Design System for Non-Axisymmetric Boundary Layer Ingestion Fans. In Proceedings of the ASME Turbo Expo 2018, Oslo, Norway, 11–15 June 2018.
78. Mandal, P. Design and Optimization of Boundary Layer Ingesting Propulsor. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2019.
79. Kumar, S. Non-AXisymmetric Aerodynamic Design-Optimization System with Application for Distortion Tolerant Hybrid Propulsion. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2020.
80. Krain, H. Swirling impeller flow. *J. Turbomach.* **1988**, *110*, 122–128. [CrossRef]
81. Hathaway, M.D. Laser Anemometer Measurements of the Three-Dimensional Rotor Flow Field in the NASA Low Speed Centrifugal Compressor; Technical report, NASA Lewis Research Center, Cleveland, OH, USA, ARL-TR-333; 1995.
82. Sato, K.; He, L. Effect of rotor-stator interaction on impeller performance in centrifugal compressors. *Int. J. Rotating Mach.* **1999**, *5*, 135–146. [CrossRef]
83. Mishra, S. Developing Novel Computational Fluid Dynamics Technique for Incompressible Flow and Flow Path Design of Novel Centrifugal Compressor. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2016.
84. Shaaban, A. Fluid Flow Controller. U.S. Patent US 6589013B2, 5 February 2001.
85. Muppana, S. Multi-fidelity Design and Analysis of Single Hub Multi-Rotor High Pressure Centrifugal Compressor. Master's Thesis, University of Cincinnati, Cincinnati, OH, USA, 2018.
86. Muppana, S.; Siddappaji, K.; Abdallah, S. High Pressure Novel Single Hub Multi-Rotor Centrifugal Compressor: Performance Prediction and Loss Analysis. In Proceedings of the ASME Turbo Expo, GT2019-91967, Phoenix, AZ, USA, 17–21 June 2019.
87. Guruprasad, A.H.; Raveesha, A.; Sherikar, A.; Siddappaji, K.; Turner, M.G. *Effect of Sweep and Translation on Wind Turbine Blades. Turbomachinery Flow Class Project Internal Report*; University of Cincinnati: Cincinnati, OH, USA, 2016.
88. Drela, M. Xfoil. Available online: http://web.mit.edu/drela/public/web/xfoil/ (accessed on 24 December 2013).